A HYBRID YOLO–EFFICIENTNET FRAMEWORK WITH RAG-BASED INTELLIGENCE FOR PERSONALIZED DENTAL DIAGNOSIS AND CARE RECOMMENDATIONS.

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CERTIFICATE

This is to certify that this project report titled **"A HYBRID YOLO–EFFICIENTNET FRAMEWORK WITH RAG-BASED INTELLIGENCE FOR PERSONALIZED DENTAL DIAGNOSIS AND CARE RECOMMENDATIONS"** is a bonafide record of work done by **CHARAN SAI CHINTA (238W5A1203), TEJOSH TELLA (228W1A1260)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

S.No	Item	Description
1	Project Title	A Hybrid YOLO-EfficientNet Framework with RAG-Based Intelligence for Personalized Dental Diagnosis and Care Recommendations.
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4	Research Group	Computer Vision, Deep Learning, and Retrieval-Augmented Intelligent Systems
5	Application Area	Health Care
6	Aim of the Project	To develop an intelligent system for dental disease detection, analysis, and personalized recommendations using YOLO, EfficientNet, and RAG
7	Project Outcomes	• Accurate dental disease detection using YOLOv8 • Efficient classification and severity analysis using EfficientNet • Personalized dental recommendations using RAG

Student Signatures

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Signature of the Guide

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LIST OF EQUATIONS

Equation	Brief Description
(1)	Input Image Representation
(2)	Detection Output
(3)	Bounding Box Coordinates
(4)	Intersection over Union (IoU)
(5)	Total Detection Loss
(6)	Confidence Score
(7)	Softmax Class Probability
(8)	Precision
(9)	Recall
(10)	F1-Score
(11)	Affected Region Area
(12)	Disease Progression (Area Change)
(13)	Confidence Score Variation
(14)	YOLO Detection Output
(15)	Feature Representation (ROI)
(16)	Classification Output (Softmax)
(17)	Categorical Cross-Entropy Loss

Equation	Brief Description
(18)	Severity Estimation
(19)	Detected Conditions Set
(20)	Knowledge Retrieval
(21)	Recommendation Generation
(22)	Cosine Similarity

ABSTRACT

Early prevention of oral health complications depends on early diagnosis of dental diseases; nevertheless, access to professional diagnostics is still restricted. The paper outlines a hybrid deep learning architecture to detect dental diseases and recommend individualized care based on intra-oral images. The proposed system combines YOLOv8 to detect and localize the object and EfficientNet to classify and analyze the severity of dental conditions. The YOLOv8 model identifies and localizes multiple dental anomalies, including calculus, dental caries, gingivitis, mouth ulcers, tooth discoloration, and hypodontia, by generating bounding boxes with confidence scores. An EfficientNet-based classifier is used to further process the identified regions of interest to enhance the robustness of classification and assist in estimating severity. Also, the system is designed to include the disease progression tracking system through a comparison of historical and the current results. A Retrieval-Augmented Generation (RAG) module is incorporated to support clinical decisions, based on a local large language model and knowledge retrieval using FAISS. This module creates customized suggestions, treatment plans, and day-to-day dental hygiene plans with the help of identified conditions and user lifestyle information. Experimental findings indicate that the proposed framework has a total accuracy of 94 and a balanced precision, recall, and F1-score of 0.94, which shows a strong and stable performance in various dental conditions. The system is implemented as a Flask-based web interface, which provides real-time analysis and interaction with users. The proposed approach provides an effective, scalable, and intelligent solution for early diagnosis, continuous monitoring, and personalized dental healthcare.

Keywords: *Dental Disease Detection, YOLOv8, Efficient Net, Intra-Oral Images, Deep Learning, Disease Classification, Personalized Recommendations, RAG, Healthcare System, Disease Tracking.*

CHAPTER – 1

INTRODUCTION

This chapter gives the background and the drive to the project, the background concepts necessary to comprehend the system, the problem statement, the impacts of the proposed work to the society, practical uses as well as a conclusion.

1.1 Origin of the Problem

The problem, which has been brought up in this project is based on the fact that the problem of early and accurate diagnosis of dental diseases is increasingly becoming an issue in a real life healthcare facility. Oral health is a big part of general health even as most of the population have untreated diseases of the mouth owing to inaccessibility of professional care, ignorance and late disease identification[1]. Common dental issues such as calculus, dental caries, gingivitis, mouth ulcers, tooth discoloration and hypodontia can have minor symptoms, but can cause serious complications in the event of their early development. This highlights the need to have good quality and accessible diagnostic solutions.

Conventionally, the manual examination is the method that was used to detect dental disease; dentists used the intra-oral images and clinical observation. This is effective yet time consuming, is subjective and to a great extent relies on the experience of the practitioner. The disparity in the lighting, the quality of the images and how they are interpreted by a human being can affect diagnosis[2]. In addition, most of the under-served communities and rural setting lack access to qualified dental care and this complicates the early diagnosis and treatment. These restrictions underscore the need to have automated mechanisms that can help in proper and consistent diagnosis.

The other significant issue is that it was possible to detect several dental problems using a single image. It is also a norm in clinical practice that there is an overlapping or co-occurring abnormality in intra-oral images and it is difficult to identify these abnormalities manually and is prone to errors. [3]Existing systems are often only image classification, in which an image is given one label, which is not adequate to deal with multiple conditions at a time. This leads to the need of advanced object detection algorithms that are able to locate and identify various dental issues in the same image with high precision[4].

Moreover, the majority of the available diagnostic tools stop at the diagnosis of the disease and it does not give any practical advice to patients[5]. Even having a diagnosed condition, patients may be not aware of preventive measures or treatment procedures. This puts a disconnect between diagnosis and decision-making. To solve these issues, the latest versions of deep learning algorithms, including YOLOv8 to detect objects of various classes and implement it with a Retrieval-Augmented Generation (RAG) module to give personal recommendations, will be used in the proposed project. This will seek to develop a smart system to identify and more so recommend the user on how to care about their oral health giving them useful tips.

1.2 Basic definitions and Background

Dental Disease Detection: Dental disease detection is a method of detecting abnormalities in intra-oral images automatically using computer vision. Automated systems are able to provide consistent results and analyze images at a faster rate than manual diagnosis. Detection of dental diseases is conducted in the proposed project to detect various conditions like calculus, dental caries, gingivitis, mouth ulcers, tooth discoloration, and hypodontia using uploaded images.

Oral Health Analysis : Oral health analysis can be defined as the examination and assessment of dental conditions found in the mouth. It entails the recognition of various kinds of diseases and how serious they are. The suggested system facilitates this process by examining intra-oral photographs and giving insightful information on diagnosed conditions.

Object Detection : Object detection is a computer vision task in which the system determines the existence of objects in a picture and their positioning. Object detection is also capable of identifying more than one region of interest at the same time as compared to image classification that assigns one label to the entire image. This renders it very applicable in dental cases where two or more diseases can be present in one image.

Bounding Box : The bounding box is a rectangle that is drawn around a dental condition that is detected in an image. It points to an exact position of the abnormality and serves to distinguish between many regions detected. The proposed system employs bounding boxes to visually identify dental problems.

Class Label : A class label is the type that a model classifies a detected object into. In this project, every identified area is identified as a particular dental disease that includes calculus, dental caries, gingivitis, mouth ulcer, tooth discoloration, or hypodontia. These labels are further used for generating recommendations.

Confidence Score : A confidence score is a measure of how sure the model is to be correct in predicting a specific class. The larger the values, the more confidence one has in the prediction. The proposed system shows the confidence scores and the results identified to assist the users in the reliability of the predictions.

YOLOv8 : YOLOv8 (You Only Look Once version 8) is a deep learning based object detection model, which is applied in the analysis of real-time images. It does localization and classification within one step, thus being quick and efficient. YOLOv8 is the main model applied in this project to identify various dental conditions in intra-oral images.

EfficientNet : EfficientNet is a convolutional neural network architecture that is developed to classify images efficiently and accurately. It sizes models in an efficient manner to realize enhanced performance. EfficientNet is applied to the proposed system to enhance the accuracy of the classification and facilitate the analysis of identified dental conditions.

Multi-Class Detection : Multi-class detection Multi-class detection is a property of a model to detect and classify objects of more than one category in a single image. The proposed system is also trained to identify a variety of dental diseases at a single time to make sure that it can be applied in real-life situations where various conditions can co-exist.

Dataset Annotation : Annotating data The procedure of labeling pictures to educate a machine learning model is known as dataset annotation. Object detection tasks are annotated with class labels and bounding box coordinates of each object. The data employed in this project is annotated in YOLO format, which allows the model to learn both localization and classification.

Training, Validation, and Testing : These are the three key phases of developing a model. The model is trained on the training set, the model performance is tuned using

the validation set and the model is tested using the testing set. In this project, the data is split into 70 percent training, 20 percent validation and 10 percent testing to effectively develop the model.

Deep Learning : Deep learning is a subfield of artificial intelligence, wherein multi-layer neural networks are utilized to acquire patterns via data learning. It has been very successful in medical image analysis. In this project, deep learning allows the system to automatically extract the features of intra-oral images and accurately diagnose the dental diseases.

Retrieval-Augmented Generation (RAG) : Retrieval-Augmented Generation is a method which involves the combination of information retrieval and language generation. Once the system identifies the presence of dental diseases, it retrieves appropriate information in a knowledge base and comes up with meaningful recommendations to the user.

Knowledge Base : Knowledge base refers to an organized body of dental knowledge, such as treatment, prevention and oral care instructions. This knowledge is then utilized by the RAG module to produce recommendations that are correct and context sensitive.

Dental Care Guidance : Dental care guidance is the step-by-step advice on how to deal with identified dental conditions. This can involve cleaning habit, treatment recommendations, dietary recommendations and preventive actions. This attribute makes the system more practical as it helps the users more than just to detect.

1.3 Problem Statement with Objectives and Outcomes

1.3.1 Problem Statement

Current dental care set-ups mainly depend on manual analysis of intra-oral scans and clinical findings in the identification of oral diseases. Despite the fact that this traditional method has worked over the years, it is usually time consuming, subjective and relies on the experience of the dental practitioner. In practical situations, the differences in the quality of the images, the light conditions, and human perception may result in inconsistent and even erroneous diagnosis. Furthermore, the scarcity of

qualified dental practitioners in rural and underserved areas also contributes to the delay in the early detection and treatment of oral diseases.

The complexity of intra-oral images is also another significant issue as several dental conditions can be present in a single image. The traditional methods of classifying images are not adequate to deal with such cases since they will tag the image using a single label without indicating the exact position of each condition. This is a drawback that makes it hard to precisely identify and identify various deviations like dental caries, gingivitis or tooth stains in various areas. Thus, a more sophisticated system is required that is capable of doing multi-class and multi-object detection with a high degree of accuracy.

Besides the issues of detection, most of the current systems are failing in offering meaning guidance and recommendations to the users, as they only emphasize on detection of dental diseases. Even in cases where a condition is identified, patients might not possess the knowledge necessary to prevent the condition or seek early treatment. This leaves a gap between the identification of a disease and effective decision-making, making the practical use of such systems less useful in real-world healthcare-use situations.

Moreover, much of the previous machine learning and deep learning methods are trained on simplified data and may not work well in the real world where images are complicated, noisy, or may have overlapping features. These constraints have underscored the importance of a powerful, scalable, and smart system that has the ability to identify various dental diseases with proper accuracy using intra-oral images besides offering personalized guidance. This project will hope to tackle these issues by incorporating an object detection model using the YOLOv8 framework with a Retrieval-Augmented Generation (RAG) module to develop an all-encompassing and viable dental healthcare assistance system.

1.3.2 Objectives

The main objective of this project is to develop an intelligent and automated dental disease detection system capable of accurately analyzing intra-oral images and assisting users with meaningful healthcare insights. The system combines the modern deep learning and knowledge-based methods of recommendation to reduce the gap between

the disease identification and the real decision-making process, and enhance the early diagnostics and prevention of dental diseases.

1. Develop a YOLOv8-based multi-class detection model to accurately identify and localize multiple dental conditions in a single intra-oral image.
2. Enhance classification performance using EfficientNet to improve feature extraction and enable detailed analysis of detected dental diseases.
3. Design a user-friendly web-based interface that allows real-time image upload, visualization, and interpretation of detection results.
4. Incorporate a recommendation engine based on RAG to produce individualized dental care recommendations, including preventive and treatment plans.
5. Introduce disease progression tracking to see the development of dental conditions over time with comparison of historical data.
6. Build a scalable and efficient healthcare system that can be extended for real-time applications and remote medical support.

Overall, the objectives aim to create a comprehensive system that not only performs accurate detection and analysis but also provides practical recommendations, contributing to improved oral health awareness and early intervention.

1.3.3 Outcomes of the Project

The successful implementation of the proposed system results in several key outcomes that demonstrate its effectiveness in dental disease detection and healthcare support. These results show how the system meets its goals through the integration of deep learning methods and practical usability aspects to apply to real-life problems.

1. Created a multi-class detecter based on YOLOv8 that can detect and locate various dental diseases in a single picture.
2. Obtained higher classification accuracy with the help of EfficientNet, which allows extracting features better and analyze identified conditions in detail.
3. Developed an easy-to-use web interface that enables users to upload images in real time and visualize the results of the detection with bounding boxes, labels, and confidence scores.
4. Implemented a recommendation system based on RAG to give individualized dental care advice, such as prevention and treatment recommendations.

5. Introduced disease progression monitoring, which allowed to compare the previous and current outcomes in order to monitor the changes in oral health.
6. Created an effective and scalable dental healthcare system that can be used in real-time and remote healthcare support.

Overall, the outcomes confirm that the system not only performs accurate detection and classification but also provides meaningful insights and guidance, making it a comprehensive solution for intelligent dental healthcare.

1.4 Related Applications

The potential applications of the proposed Dental Disease Detection and Recommendation System to real-life healthcare and other areas with a high need of early diagnosis and preventive care are quite numerous. The main use is in clinical decision support systems where dentists can use the model to help in the diagnosis of the dental conditions based on intra-oral images. The system can be used to offer supportive functionality as the results of quick and accurate detection can be presented to assist professionals to save time in diagnostic procedures and enhance consistency in clinical judgments.

The other significant use is tele-dentistry and remote healthcare services. Dental professionals are not readily available in most rural or underserved areas. This system is capable of fitting into telemedicine systems, where the patients will be able to post their intra-oral images and get an initial analysis and instruction. This does not only enhance access to dental care but also early detection and timely intervention, minimizing the chances of life threatening complications.

The system is also very applicable in dental education and training. The application provides students and trainees with an opportunity to learn about different dental conditions and their visual detection outcomes and explanations. Through the interpretation of various intra-oral images and predictions, learners are able to enhance their diagnostic skills and have a hands-on experience of a real-world case in a well-organized way.

Besides, the project can be implemented in preventive health care and awareness systems. The system promotes individuals to have better oral health by offering

personalized recommendations like oral hygiene, dietary, and early treatment. This may be applied in mobile or web based apps to create awareness on the importance of having dental care and to make the users engage in preventive measures before the situations deteriorate.

The other important use is in the health monitoring and record management systems where the idea of disease progression can be applied. With the help of storing the results of past analysis and comparing them to the present detections, the system can assist in tracking the state of dental conditions improvement or deterioration over a certain period. This characteristic can help patients and healthcare professionals to measure the success of treatment and design the future care plans.

On the whole, the proposed system can be applied to academic and practical settings (clinics, telemedicine platform, educational tools, and personal healthcare applications) and is a worthy addition to the modern technology in dental and healthcare fields.

CHAPTER – 2

REVIEW OF LITERATURE

2.1 Description of Existing Systems

In [1], deep learning was used to classify dental panoramic radiographs on the basis of age groups. The experiment proved that the convolutional neural networks can be used to extract discriminative features of the dental images and obtain the high level of reliability in the classification process. Nevertheless, it was restricted to classification and failed to deal with multi-condition detection and localization of abnormalities in the teeth.

In [2], a superior architecture of a transformer-based YOLO framework was suggested to optimize on a genetic algorithm and detect and classify dental anomalies. The model was more accurate and efficient at recognizing types of tooth and conditions of the dentures. This article emphasises the utility of contemporary object detection methods, although it mainly addresses the performance of detection and lacks decision-support systems.

A deep learning system that is based on a DenseNet model was proposed in [3] to diagnose oral diseases. The model had high classification accuracy in terms of learning complex visual patterns established over dental images. Although effective, the system could only classify single labels and it could not detect and localize multiple objects in real-time.

A hybrid deep learning structure based on metadata retrieval was introduced in [4] to detect, classify and predict. The experiment has shown that a combination of several deep learning methods can enhance the performance and offer more in-depth insights. Nevertheless, the system was not particularly developed to analyze dental images and did not target intra-oral multi-class detection.

A system that combines visual diagnosis and Retrieval-Augmented Generation (RAG) was suggested in [5] in the treatment of dermatological diseases. The study demonstrated that structured and reliable recommendations can be obtained by integrating retrieval-based knowledge and language models. Though this method

illustrates the potential of RAG in healthcare, it is not directly applied to detect dental disease based on images.

In [6], dental disease classification using intra-oral images was done using deep learning models. The paper showed that neural networks can be successfully trained to learn visual characteristics and diagnose oral conditions. Nevertheless, the method was confined to classification and lacked the object localization or multi-condition classification.

In [7], they applied a computer vision-based approach to detect and classify dental caries in intra-oral pictures. The findings revealed that AI systems have the potential to enhance the accuracy of diagnosis over manual inspection. Nevertheless, the system was limited to one disease and failed to treat various cases of dental conditions.

A deep learning-based method to identify oral abnormalities in clinical images was suggested in [8]. The model worked well in controlled situations but had difficulties when it was implemented in real life situations where overlapping or complex conditions would be involved.

A convolutional neural network-based image recognition system was created in [9] to identify dental diseases and provide preventive healthcare. The research highlighted the significance of automated diagnosis to minimise human error, yet the system did not have a real-time detection capacity and multi-object recognition.

In [10], dental image classification and analysis were done with a deep convolutional neural network. The results showed that CNN-based models can significantly improve diagnostic accuracy compared to traditional approaches. Nevertheless, like other classification-based systems, the system lacked object location and customized suggestions.

In [11], models based on EfficientNet were used to detect dental diseases and showed better rates of accuracy and computational efficiency than the conventional CNN architectures. The significance of optimized architectures on improving performance was emphasized in the study, which was still restricted to classification tasks.

YOLOv7 was applied in [12] to detect dental caries, especially to cope with the issues of small objects. The researchers revealed that the object detection models were capable of detecting fine-grained dental abnormalities, but they had to be refined to enhance accuracy on the more difficult cases.

A comparative analysis between YOLOv7 and YOLOv8 was performed in [13] to detect dental caries with the help of intra-oral images. The findings showed that YOLOv8 is more accurate, faster inference, and has a better detection performance, so it is more adaptable to real-time dental applications.

In [14], the authors suggested ways of improving YOLO-based models with attention mechanisms and additional heads of detection as potential ways of improving the performance of detection. These changes resulted in greater precision in the diagnosis of dental diseases, particularly in complicated and intersecting cases.

In [15], more sophisticated clinical decision support systems that combine large language models with Retrieval-Augmented Generation were studied. The research showed that knowledge retrieval combined with AI-based detection can yield correct and context-specific recommendations and minimize the amount of hallucinations in medical AI systems.

A YOLOv8-based model is applied in detecting interproximal caries using bitewing radiographs in [16]. The analysis shows enhanced image clarity and decreased distortion, which allows correct classification of the image into the categories of enamel, dentin, and restoration. The model achieves good precision and recall, showing its effectiveness for early-stage dental diagnosis.

The work in [17] proposes a YOLOv9-based framework to enhance automated dental diagnostics. It introduces improved feature extraction and deeper network architecture, resulting in better detection of subtle abnormalities. The model shows around 22% improvement in accuracy, indicating the impact of architectural advancements.

A real-time detection system using YOLOv5 is presented in [18], where both X-ray and color dental images are analyzed. The model achieves approximately 90% accuracy and benefits from preprocessing techniques like the YIQ color model, which improves image quality and detection performance.

In [19], YOLOv5 is applied for detecting dental diseases and corresponding treatments using panoramic X-ray images. Although the model uses a rather small dataset, the model is very precise and has a decent mAP, proving to be effective in multi-class detection with appropriate annotation.

Another approach in [20] utilizes YOLOv6 for detecting dental pathologies and treatments from panoramic X-rays. The model has reached a mAP of 70.76% and demonstrates the difficulty in having large labeled datasets, but displays high potential in aiding clinical diagnosis.

In general, the literature review demonstrates that great advances have been achieved in the application of deep learning methods to detect dental diseases. But the majority of the available systems are either classification-based or detection-based and lack a holistic approach to detecting and making recommendations based on the accurate multi-class detection. The limitations presented in the proposed project are overcome in a way that YOLOv8-based object detection is combined with a Retrieval-Augmented Generation module which provides both accurate diagnosis and valuable dental care advice.

2.2 Summary of Literature Study

The literature review indicates that a slow shift in dental image analysis methods towards more sophisticated deep learning methods and real-time object detection systems has been observed. It has been shown by previous researchers that deep learning networks, especially convolutional neural networks are very effective at learning complicated visual patterns on dental images and greatly outperform the traditional methods in making a diagnosis [6]. This progress has been enhanced with the application of transfer learning, hybrid models, and optimized CNN models, which improve the performance of classification and robustness in different datasets and under different imaging conditions [9][10].

The acceptance of object detection models based on the use of YOLA is increasing in recent research, and it is one of the most important trends in this area. The models can detect and localize several abnormalities in a single intra-oral image and therefore are more applicable in real-life situations than the conventional classification methods [3]. Recent advances in YOLO architectures have also increased the speed, accuracy, and

efficiency of detection making them capable of being used in real-time healthcare systems [4]. Also, research has revealed that the models are reliable even in difficult environments like changes in lighting, noise and quality of images [1].

Besides these developments, other more recent works have investigated more advanced versions of YOLO architectures to achieve better diagnostic performance. An example is Yolov8, which is shown to have a high potential of identifying premature dental caries in bitewing radiographs, with better accuracy and categorizing the levels of disease severity [16]. On the same note, the introduction of YOLOv9 has also improved detection performance with improved feature extraction and more network structures, which have translated to significant improvements in accuracy and sensitivity [17]. Previous frameworks like YOLOv5 have also been extensively used to identify dental issues in real-time, with high accuracy and computational efficiency [18]. Moreover, YOLO-based models have been effectively adapted to identify dental disease and dental treatment based on panoramic X-rays, even with relatively small datasets [19]. The benefits of deep learning models in detecting several dental pathologies and treatment conditions with the use of YOLOv6 have been validated in other studies, though they also emphasize the problem of annotated datasets scarcity in medical imaging [20].

The other interesting finding in the literature is that the majority of systems currently available are mainly concerned with the accuracy of the detection and classification of objects with minimal attention paid to the user assistance after detection. In spite of the fact that these systems give outputs, including class labels, confidence scores, and bounding boxes, they tend to have no ways to advise users on correct preventive measures or treatment options [8]. This results in a disconnect between the technical diagnosis and field practice of healthcare provision and makes the overall usability of such systems less practical in the real environment.

Recent research has also highlighted the potential of Retrieval-Augmented Generation (RAG) and large language models for generating structured and context-aware recommendations based on curated knowledge sources. These methods mix information retrieval and natural language generation to provide useful and credible advice [5]. Although the identified methods are not fully implemented into dental image-based detection systems so far, they are a promising pathway towards improving intelligent healthcare solutions to fill the gap between diagnosis and decision-making.

CHAPTER – 3

PROPOSED METHODOLOGY

3.1 Design Methodology

The approach to the design of the suggested system lies in the creation of a multi-class dental disease detection and multi-user customized recommendation generation integrated into a single workflow. The approach will help counter the shortcomings of the traditional dental diagnosis frameworks, where diagnosis is carried out manually and does not provide practical recommendations to the consumers. The proposed system combines a YOLOv8-based object detector model with an EfficientNet-based classification model and a Retrieval-Augmented Generation (RAG) system to overcome these issues. This will enable the system to detect dental diseases using intra-oral images and give meaningful healthcare advice using a formal body of knowledge.

Data collection and preparation will be the beginning of the methodology. In the case of the visual detection part, a set of intra-dental images is employed which is annotated with the YOLO format. Data set is composed of a range of dental diseases such as calculus, dental caries, gingivitis, mouth ulcers, tooth discoloration and hypodontia. The system also does the object detection, thus a single dental image can have multiple dental abnormalities with each being represented by a bounding box and the respective class label. At the same time, a database of dental care instructions, treatment regimens and prophylaxis are prepared. The system is based on this two-data scheme where image-based learning is used to detect and text-based knowledge learning is used to generate recommendations.

The preprocessing and data organization come after data collection. Images are all resized to a standard resolution (e.g., 640 × 640 pixels) to fit the input of the YOLOv8 model. These data are then divided into the training, validation and testing sets in ratio of 70:20:10 in order to have systematic development and testing of the model. Horizontal flipping, scaling, rotation, and brightness adjustments are data augmentation techniques used in enhancing the model robustness and minimizing overfitting. These pre-processings come in handy with regard to generalizing the model to image quality variations, orientation and lighting variations.

The system is handed over to the detection stage of the YOLOv8 after preprocessing. The intra-oral image is processed by the trained YOLOv8 model identifying objects in a single-stage pipeline. The model identifies various dental problems in the image and generates bounding boxes, labeling classes, and confidence values. YOLOv8 is chosen because of its compromise between accuracy and real-time inferences. The raw input images are converted to structured detection results by the output of this stage and used in the other stages of the system.

Once it is detected, the methodology includes enhancement of classification using EfficientNet. The resulting regions of interest are further narrowed down to improve classification accuracy and aid in the analysis of dental conditions in detail. The predictions are refined using EfficientNet and the predictions obtain deeper feature representations, which are more predictive of diseases and provide insight into the severity.

Multi-object label extraction and organization is the second step. The system considers all the class labels estimated of the regions which are identified and classifies them into structured data. As there can be several dental conditions in one image, each detected region is handled separately. This ensures that all the diseases detected are factored and further analysis and recommendations are produced.

After the extracted and identified labels, the system proceeds to the knowledge retrieval process basing on the RAG framework. The dental conditions identified are used as queries to locate the appropriate information in the knowledge base prepared. The retrieval system retrieves text snippets regarding treatment procedures, preventive care and oral hygiene practices and risk factors. This will make sure that the recommendations will be founded on facts and condition-specific knowledge as opposed to general text generation.

A large language model then processes the retrieved information to give structured and user-friendly recommendations. The model organizes the data into step-by-step instructions of each of the identified conditions that can be easily understood. Such recommendations can consist of cleaning habits, dietary recommendations, advice on medical consultation, and prevention. This will transform raw detection outputs into effective healthcare support that will render the system more realistic and easier to use.

The last part of the methodology is user interface and presentation of results. The entire system will be implemented into a web-based application (such as Flask or Streamlit) where the users can post intra-oral pictures. The system displays the original image, the bounding boxes located, the predicted labels and the scores of the confidence. Also, the recommendations obtained are in a structured format. This interface offers a user friendly interface which is easy to use as well as the interface can be easily accessed by non technical users.

On the system level, the suggested methodology may be considered a series of modules that are interrelated: dataset preparation, preprocessing, YOLOv8 detection, EfficientNet classification, label extraction, knowledge retrieval, recommendation generation, and user interface presentation. The components are combined to produce a fully-fledged and smart dental healthcare support system that combines the correct identification with valuable advice.

3.2 System Architecture Diagram

The proposed framework is a multi-class dental disease detection, severity analysis, disease tracking, and personalized recommendation generation integrated system architecture proposal. However, this architecture ensures the combination of deep learning, computer vision, and knowledge-based intelligence into one pipeline to deliver the correct diagnosis and valuable healthcare advice, unlike traditional systems that prioritize one of these methods.

The system has two significant stages as revealed in Figure 3.1, the model processing stage, and the user interaction stage. The system is trained in its first stage with annotated intra-oral image datasets. The training sets are split into training, validation, and testing sets to make sure that the learning and performance evaluation are appropriate. It uses a hybrid scheme of YOLOv8 and EfficientNet in which the former is used to detect objects in multiple classes and the latter to classify them and improve its classification by using more and meaningful features. This stage preconditions the proper and credible detection of dental diseases.

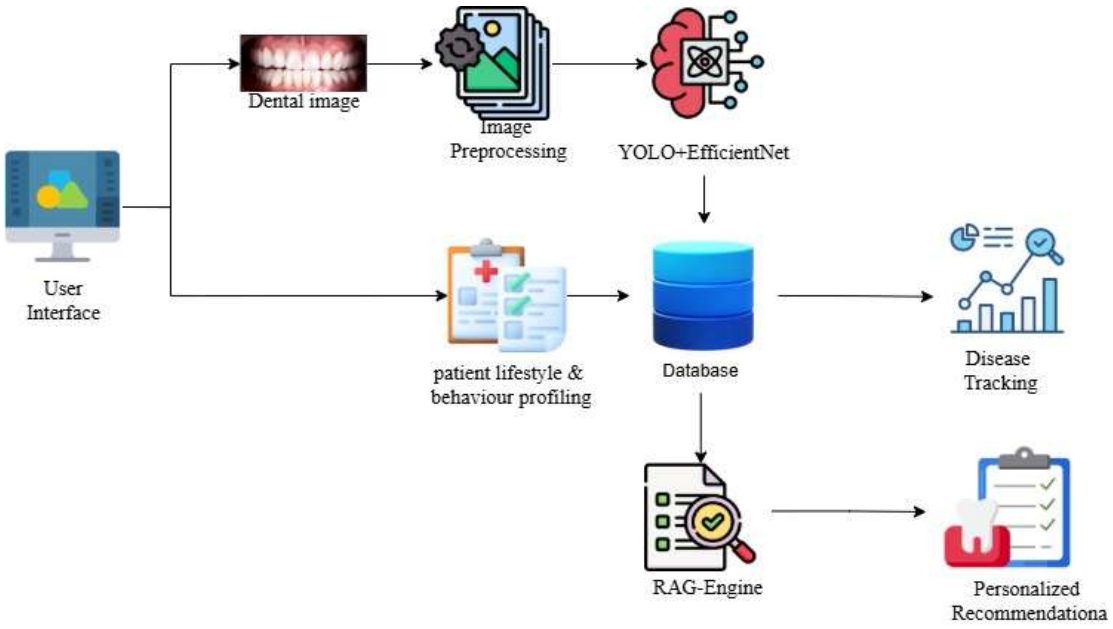

Figure 3.1: Architecture diagram of the Dental Disease Detection system.

The system runs in real-time, after training, via a user interface. This will start with the user uploading an intra-oral image, which is subjected to preprocessing, including resizing, normalization, and image enhancement to enhance the image quality and compatibility with the model. The processed image is subsequently sent to the hybrid YOLOv8-EfficientNet model, which identifies several dental conditions at once and creates bounding boxes, labels and classifications, and scores.

The outcomes of the detection are saved in a database containing patient records, prediction results, and history. It allows the system to facilitate a disease tracking system, in which previous and present outcomes are assessed to keep track of the development of dental conditions over time. This kind of tracking is useful in determining the progress or decline in teeth health.

In order to be more usable, the system has a Retrieval-Augmented Generation (RAG) module to generate personalized recommendations. Depending on the circumstances detected and the information about the patient, the system accesses pertinent information in a structured form of knowledge base of dental care guidelines and treatment procedures. A language generation part then takes this information and generates clear and actionable recommendations, such as preventive measures, oral hygiene practices, and treatment recommendations.

Lastly, the findings are provided in a user-friendly format where the user can see identified areas, forecasted tags, confidence scores, and customized suggestions in an organized design. The combination of the detection, tracking, and recommendation modules will make sure that the system is not only able to detect dental diseases but also offers continuous monitoring and practical healthcare assistance.

3.3 Description of Algorithms

The dental disease detection system proposed is created by a combination of the advanced deep learning algorithms to detect, classify, analyze the severity and generate recommendations to be accurate. The system is a combination of object identification models, convolutional neural networks, and knowledge-based solutions to develop a scalable and efficient healthcare solution.

In comparison to the single-model methods that were used traditionally, the system uses a hybrid architecture in which object detection algorithms are used to localize areas of affected areas in dental images, and classification models are used to analyze features and the severity of the identified conditions. This multi-step method enhances accuracy and reliability.

Moreover, the system includes the time-based analysis that will monitor the evolution of the disease by comparing the past and current outcomes to monitor oral health continuously. It also contains a Retrieval-Augmented Generation (RAG) module to create individual recommendations depending on the identified conditions and information about the knowledge base.

On the whole, the framework is a holistic system incorporating detection, analysis, tracking, and recommendation, and is applicable to real-life dental healthcare settings.

3.3.1 YOLOv8m

YOLOv8 (You Only Look Once version 8) is a state-of-the-art object detector algorithm that is popular in real-time detection applications because of its speed and accuracy. YOLOv8 is used to identify the different types of dental conditions in panoramic X-ray images and intra-oral images in the proposed system. The model can detect many dental anomalies including cavities, fillings, impacted teeth, implants,

caries, calculus, gingivitis, ulcer, tooth discoloration, and hypodontia in only a single forward pass.

The critical strength of YOLOv8 is that it is designed as a single-stage detector, that is, localization and classification are done simultaneously. This removes the need to perform individual region proposal steps and is much more efficient computationally. The model works on the whole picture simultaneously and is able to learn the spatial relations between various parts of the image and identify several objects with a high degree of accuracy.

The input image is represented as:

$$I \in R^{640\times640\times3} \tag{1}$$

where I denotes the downsampled RGB dental picture.

The model produces detect outputs as:

$$D = \{(b_i, c_i, s_i)\}_{i=1}^{N} \tag{2}$$

where b_i is the bounding box, c_i is the predicted class, s_i is the confidence score, and N is the amount of objects detected.

A bounding box of each one is:

$$b_i = (x_i, y_i, w_i, h_i) \tag{3}$$

where (x_i, y_i) is the center coordinates, and w_i, h_i represent width and height.

The localization accuracy is measured by the Intersection over Union (IoU) metric:

$$IoU = \frac{Area(B_p \cap B_{gt})}{Area(B_p \cup B_{gt})} \tag{4}$$

The detection loss function is:

$$L_{total} = L_{box} + L_{cls} + L_{obj} \tag{5}$$

where L_{box}represents localization loss, L_{cls}represents classification loss, and L_{obj}represents objectness loss.

The confidence score is computed as:

$$Conf = P(Object) \times IoU \tag{6}$$

Conf in (6) is the confidence of the detection, P(Object) is the probability that an object is contained in the bounding box and IoU is the overlap between predicted and ground truth boxes.

The class probability is calculated using the Softmax function:

$$P(c_i) = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}} \tag{7}$$

In (7), $P(c_i)$is the probability of class i, $z_i$is the raw score for class i, and $C$is the total number of classes.

The metrics of evaluation are: precision, recall, and F1-score:

$$Precision = \frac{TP}{TP+FP} \tag{8}$$

In (8), Precision is used to determine accuracy of positive predictions, where TP are true positives and FP are false positives.

$$Recall = \frac{TP}{TP+FN} \tag{9}$$

In (9), Recall measures the ability to detect all actual positives, where FN are false negatives.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{10}$$

F1-score in (10) is a harmonic mean between precision and recall, indicating a balance in evaluation measure.

This model enables efficient and real-time detection of multiple dental conditions with high accuracy.

3.3.2 YOLOV8n

The results of the detection in the YOLO model are further used to monitor the development of dental diseases with time. The ability to track the progression of the

disease is also a significant part of the system because it enables constant monitoring of the patient conditions and determines whether a disease is getting better, worse or not changing.

The results of the detection by YOLO are further used in disease progression analysis through comparing the traits of the bounding box with time. The size of the region identified is calculated as:

$$A_t = w_t \times h_t \tag{11}$$

where A_t represents the affected region at time t.

Disease progression is determined by comparing the variation in area across different time instances:

$$\Delta A = A_{t2} - A_{t1} \tag{12}$$

An increase in area indicates disease progression, while a decrease indicates recovery or improvement.

The variation of confidence over time is also taken into consideration:

$$\Delta s = s_{t2} - s_{t1} \tag{13}$$

where s represents the confidence score of detection.

The system classifies progression as improved, worsened or stable according to threshold values of area and confidence differences. This mechanism allows ongoing observation of dental conditions and helps to prompt alerts and preventive measures.

3.3.3 Hybrid YOLO–EfficientNet

The proposed system will use a hybrid deep learning model comprising of object detection, image classification, and knowledge-based generation of recommendations to offer a full dental diagnosis and decision-support model. In this architecture, the YOLO-based model will be used to allow precise localization of dental abnormalities, whereas EfficientNet will be used to carry out fine-grained classification and severity assessment of the localized areas. This combination helps the system to seize both spatial and semantic features, which enhance the overall output of the system in terms of diagnostic performance. The detection model is concerned with the presence and

location of dental conditions, and the classification model is concerned with extracting more detailed feature representations, and determining the severity and clinical importance of each condition.

YOLO model first identifies areas of interest:

$$D = \{(b_i, c_i, s_i)\} \tag{14}$$

Regions found are extracted and sent to the EfficientNet model to extract features and classify the severity. The feature representation is:

$$F = f(I_{ROI}) \tag{15}$$

where I_{ROI}represents the cropped region of interest.

The classification output is obtained using Softmax:

$$y = Softmax(WF + b) \tag{16}$$

where y represents the predicted severity class.

The classification loss is defined using categorical cross-entropy:

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{17}$$

Severity estimation combines bounding box area, classification output, and detection confidence:

$$S = \alpha A + \beta s + \gamma y \tag{18}$$

where $\alpha, \beta, \gamma$are weighting parameters.

The detected conditions are represented as:

$$C = \{c_1, c_2, \ldots, c_m\} \tag{19}$$

These conditions are used as queries for the Retrieval-Augmented Generation module. The retrieval system fetches relevant knowledge $R_j$from a database:

$$R_j = Retrieval(c_j) \tag{20}$$

The final recommendations are generated as:

$$G_j = f(c_j, R_j) \tag{21}$$

For embedding-based retrieval, cosine similarity is used:

$$Similarity = \frac{q \cdot d}{\|q\|\|d\|} \tag{22}$$

where $q$is the query vector and $d$is the document vector.

This combined solution facilitates the creation of recommendations that are context-sensitive based on visual recognition, incorporating medical knowledge and patient-specific data.

The integration of YOLO-based detection, EfficientNet-based classification and RAG-based recommendation generation creates a strong and smart dental healthcare system. The detection model is used to localize dental conditions with accuracy whereas the classification model is used to assess the severity. Progression tracking mechanism allows monitoring of patient health at all times. The Retrieval-Augmented Generation also enhances the system by offering personalized and context-sensitive suggestions in response to the identified conditions and in accordance with the medical knowledge that is considered. Such a hybrid framework has great potential to enhance the accuracy, interpretability, and usability of the diagnostic, and it is a scalable and practical solution to the dental healthcare application in the real world.

3.4 Description of Datasets, Requirements and Tools

A complex set of visual data, textual sources of knowledge, and sophisticated computation tools are used to create the proposed dental disease detection system, which will be able to achieve accurate detection, classification, and monitoring of disease progression and generate personalized recommendations. The proposed system, in contrast to traditional systems, that use image classification or detection only, uses a dual-data and multi-module architecture where image and knowledge-based input are combined into a single pipeline. This combination enables the system to be utilized as a diagnostic, yet an intelligent clinical assistant which can give practical insights and recommendations. The model is a combination of deep learning to detect objects (YOLOv8) and classify (EfficientNet) and a Retrieval-Augmented Generation (RAG) module to generate outputs driven by knowledge. This architecture makes sure that the

system can be used to achieve not only visual comprehension but also contextual reasoning, thus becoming more efficient and closer to real-life dental healthcare contexts.

The system is to deal with various types of dental data, both panoramic X-ray and intra-oral images, which constitute diverse dimensions of dental diagnosis. The fact that various data sources are used makes the model stronger and provides the model with the ability to generalize with other types of dental imaging conditions. Also, the incorporation of systematic body of knowledge enables the system to close the gap between crude model forecasts and useful clinical advice. The datasets, system requirements and tools adopted in this project are well chosen and combined to help each phase of the pipeline, i.e. data preprocessing and model training to inference, retrieval and user interaction.

Dental Image Dataset

The major visual data in this project are dental images obtained in publicly available repositories like Kaggle, Roboflow, and other curated data sources that offer annotated intra-oral and panoramic X-rays images. The dataset is created to encompass a broad scope of dental conditions such as caries, calculus, gingivitis, tooth discoloration, ulcers, hypodontia, cavities, fillings, impacted teeth and implants. These classes correspond to early and advanced dental cases, and the model can be trained to learn a set of features that are comprehensive and can be applicable in the real world.

As the system is grounded on object detection, annotating each image in the dataset with bounding boxes is used to describe the spatial position of the dental anomalies. Every annotation consists of a class label and the coordinates of the bounding box, allowing the YOLOv8 model to learn localization and classification at the same time. This object-level annotation is especially significant in the dental applications, since several conditions may co-exist in the same picture and may be present in various areas. The bounding box annotations enable the model to isolate and detect each condition separately, thus enhancing the accuracy of detection.

The data is provided with panoramic X-ray images as well as intra-oral images to ensure representation of various angles of the dental conditions. Panoramic X-ray gives the entire picture of the tooth structure, jaw bone and connective tissues and is therefore

appropriate to detect structural defects like impacted tooth and implants. On the contrary, intra-oral images concentrate on the surface of the teeth and gums, with the ability to identify caries, calculus, gingivitis and discoloration. Adding such a variety of types of images will make sure that the model is trained on various data and will be able to generalize to different clinical situations.

Besides basic annotation, the dataset is optimized to improve the quality and consistency of images. This involves pixel value normalization, contrast adjustment and noise denoising. Pictures are also standardized in the size and format to make them compatible with the model input requirements. These two preprocessing methods are useful in enhancing the performance of models by giving clean and consistent input data. Moreover, the data set consists of lighting variations, angles of the shot, and camera equipment, which assists the model to become learned on strong features that are insensitive to changes in the real world.

Division and sorting of Data.

The dental image dataset is categorized into training, validation and testing subsets in a systematic manner to facilitate the development of the model and unbiased assessment. A split of 70, 20 and 10 percent is used to divide the dataset into training, validation and testing. The division allows the model to be trained using a large dataset and also has different data to be used as validation and testing.

The visual features of the various dental conditions are learned using the training dataset. At this stage the model is taught to detect patterns of color changes, texture and structural anomalies. The training procedure is a series of epochs, where the model repeatedly re-optimizes its parameters to reduce the loss function. The validation dataset is used during training to monitor model performance and tune hyperparameters such as learning rate, batch size, and number of epochs. It assists in identifying overfitting and is also a good way of making sure that the model will generalize to unseen data.

Testing dataset is applied once the training process is achieved to assess the final performance of the model. As the data used in testing is not utilized in the training and validation stages, it will be an objective indicator of the accuracy and reliability of the model. Precision, recall, F1-score and mean Average Precision (mAP) are used as

performance metrics and are computed on the test data to determine the performance of the model.

Any image is scaled to a common size of 640 by 640 pixels to suit the input size of the YOLOv8 model. This is to guarantee uniformity and less complexity in computation. The training dataset is enhanced with data augmentation methods to enhance the model robustness. Such techniques are flipping, rotation, scaling, brightness, contrast, and addition of noise. Augmentation adds to the variety of the dataset, and it aids the model to work effectively across various conditions.

The data is structured in YOLO format where every picture has an annotation file with the class tags and normalized bounding box positions. This systematized organization guarantees effective training and facilitated incorporation into the YOLOv8 system.

Dental Knowledge Base

Besides the image dataset, the system has a dental knowledge base, which is the textual dataset of the Retrieval-Augmented Generation one. The knowledge base is built based on the clinical guidelines, research publications, dental textbooks, and curated healthcare resources. It has comprehensive content regarding different dental disorders, such as symptoms, causes, treatment, prevention, as well as suggested daily care procedures.

The knowledge base is organized in such a manner that it can easily retrieve the relevant information based on the conditions detected. Every dental condition is linked with various bits of information that can be taken to produce recommendations. As an illustration, a diagnosed case of gingivitis can be connected to the data concerning the oral care behavior, nutrition, and the professional treatment.

Knowledge base is indexed with the help of the vector embedding and this enables similarity search to be performed efficiently in the retrieval process. Upon the detection of a dental condition, the condition is translated to a query vector and compared to stored embeddings to find the most relevant data. This makes the recalled knowledge contextually relevant and related to the condition identified.

The knowledge base integration makes the system an intelligent assistant that does not only give detection results but also actionable insights. This is especially applicable in health care where the users need some guidance besides diagnosis.

Proposed System Requirement

The proposed system should be implemented with the help of software tools, libraries, and computational resources to serve various pipeline stages. The main programming language is Python because it is flexible and has a wide range of machine learning and deep learning frameworks.

Object detection is performed on the YOLOv8 model that allows the system to detect and localize dental conditions with high precision. Classification and severity analysis is done with EfficientNet, giving more information on anomalies identified. The models are trained in an environment like Kaggle Notebook and Google Colab that offer the use of Gpus to compute effectively.

Image processing applications using OpenCV include resizing, normalization, and visualization. It is very important in processing image data, and in integrating various components in a seamless way. Flask is also employed by the system to build the web-based interface, which enables the users to communicate with the system through posting images and results.

A database, called MongoDB, is utilized to store user data, history of detection, and progression tracking. This allows the system to keep records and facilitate temporal analysis of dental conditions.

Tools Used in the Proposed Project

The suggested project involves a complex set of tools and technologies to assist various steps of the system such as the preprocessing of data, training the model, inference, knowledge retrieval, and interaction with the user. All the tools are chosen depending on their competence to efficiently manage certain aspects of the pipeline, so that the whole system is a smooth and coherent system. YOLOv8 is the main object detector that will be used as the main base of deep learning to detect and locate dental anomalies. YOLOv8 is the choice because it has a high level of detecting, real-time, and can identify multiple objects in a single image. It takes input dental images and provides

bounding boxes and class labels and confidence values, and thus is very applicable in detecting complex dental conditions in various types of imaging (intra-oral images and panoramic X-rays).

Besides detection, EfficientNet is also employed as the severity model of detected regions. EfficientNet is characterized by its compound scaling methodology, balancing network depth, width, and resolution, allowing it to be highly accurate using a relatively small number of parameters. EfficientNet boosts the system in terms of distinguishing between different levels of severity of dental conditions by extracting high-level features within the found regions of interest. This enhances the general diagnostic power of the system as it not only delivers the result of detection but also clinically relevant information.

Image preprocessing and image manipulating are widely performed with the help of OpenCV throughout the system. It carries out functions like image reading, resizing, normalization, noise reduction and visualization of detection results. OpenCV also allows movement of image data among various modules of the system, which allows smooth passage of data between the user input and model inference stages. It is a key part of the pipeline because of its capacity to effectively process image transformations.

For building the user interaction layer, Flask is used as the web framework to develop a lightweight and responsive application interface. It allows users to post dental photos, results of detection with a bounding box, and get generated recommendations in a structured format. Flask is the intermediary between the back-end processing system and the front-end user interface to maintain a smooth flow of communication and real-time interaction.

MongoDB is employed as the database management system to store the user data, history of detection, and data on disease progression. It enables easy storage and retrieval of both structured and unstructured data, which will enable the system to keep historical records of patient conditions. This accumulated information is especially helpful in monitoring the development of the disease over the course of time and sending alerts according to the alterations in the identified conditions.

FAISS (Facebook AI Similarity Search) is used as the retrieval engine in the Retrieval-Augmented Generation module. It allows one to search the dental knowledge base

efficiently (similarity search) by embedding the textual data in vectors. Upon the identification of a dental condition, FAISS finds the most pertinent information by matching the query embedding with pre-stored document embeddings. This is to make sure that the content that is retrieved is contextually relevant and in line with the identified condition.

The recovered information is then fed to a language generation model (e.g. a local large language model or the Google Gemini API) that transforms the knowledge recovered into structured and human readable recommendations. Language model is important in converting technical medical data to comprehensible and practical advice to users. Retrieval and generation are also integrated so as to make the recommendations accurate and easy to use.

Every system is built and run within platforms like Kaggle Notebook or Google Colab that have the capability of training and experimenting using the GPUs. The platforms facilitate quicker calculation, repeatability of findings and scalability to process large data. All in all, the integration of these tools will make sure that every element of the system will work efficiently and will be part of the overall functionality of the framework.

Final Integration

The proposed project is a system that combines the datasets, system requirements, and tools in a way that creates a comprehensive end-to-end dental disease detector and recommender system. The process of integration makes sure that every component is integrated in a co-ordinated way so that there is a smooth flow of data in the input and the output. The system starts by the user loading an image of the dental on the web interface, after which, the image is resized and normalized using OpenCV. The processed image is subsequently fed into the YOLOv8 model that identifies and localizes dental anomalies by producing bounding boxes, class labels, and confidence scores.

The identified areas are cut out and sent to the EfficientNet model to be classified and analyzed in terms of severity. This action improves the results of the detection process as it offers more information concerning the state of the detected anomalies. The result of the detection and classification modules is then synthesized to create a global view

of the dental situation. The MongoDB database stores these results, enabling the system to keep historical information and facilitate tracking of disease progression.

After identifying and analyzing the conditions, the system transmits the identified class labels to the Retrieval-Augmented Generation module. With the help of FAISS, the dental knowledge base is accessed to find the information of interest according to the matching similarity. The model is then fed with the retrieved content which is then processed by the language generation model that comes up with personalized suggestions based on the identified conditions and the context of the user. This makes sure that the output is accurate not just meaningless and easy to interpret.

The Flask interface shows the final result to the user, showing the original image with the detection result and severity analysis and a personalized recommendation. This end-to-end pipeline is necessary to make sure that the system is used as an integrated solution and not a system of autonomous modules. All elements play a part in a particular task but are aligned to the overall process.

Combining datasets, tools, and algorithms makes the system scalable and intelligent to provide real-time analysis and decision support in healthcare. It connects the visual detection and clinical interpretation process by integrating knowledge-based reasoning with deep learning. Consequently, the suggested framework is not only technically efficient but it is also applicable in practice in the real-life dental healthcare applications, which allow diagnosing, monitoring and offering users with personal guidance.

CHAPTER – 4

RESULT & OBSERVATIONS

4.1 Stepwise description of Results

This part of the paper provides the findings of the proposed dental disease detection system in a tabular and organized way. The results are arranged in a chronological order according to various stages of developing the project. The initial phase is dedicated to the YOLOv8-based detection model training on panoramic X-ray images, the second phase involves the enhanced detection system on intra-oral real-life images, and the third phase is the hybrid YOLO-EfficientNet model with a Retrieval-Augmented Generation framework. This setup enables one to clearly see how the system developed into a rudimentary system of detection to a full scale intelligent dental healthcare solution.

The step-by-step representation is significant as every phase of the project is aimed at enhancing the general functionality of the system. At the first level, the emphasis is on the correct detection and localization of dental abnormalities. Later functionalities including severity analysis, disease progression and generation of personalized recommendations are added in the later stages. The development of this system is indicative of the way the system is evolved over time into a complex clinical decision-support system.

Moreover, the findings are represented in graphical forms like accuracy curves and loss curves, and tabular summaries. The visualizations are useful in determining the learning behavior of the model in training and validation. Graphs can be used to analyze convergence trends, training stability, and the capacity to generalize whereas tables give a brief overview of the main performance metrics. Such presentation forms a structure that makes it clear and assists to examine the efficiency of every algorithm involved in the system.

The findings also show the gains made at every level in the accuracy, reliability, and the usability. The initial phase shows that YOLOv8 is able to identify dental anomalies based on structured X-ray images and the subsequent steps go further to show that the

same is applicable to actual-life situations and incorporates knowledge-based reasoning. On the whole, this section gives a detailed discussion of the performance of the proposed system and the way in which its performance changes as more advanced components are introduced.

4.1.1 YOLOv8-Based Dental Detection Results

The initial step of the suggested system is dedicated to the usage of the YOLOv8 model in detecting dental anomalies on the panoramic X-ray pictures. This step forms the basis of the whole framework since it develops the ability of deep learning-based object detection in detecting and locating dental diseases. The aim of this phase was to eliminate the manual system of inspection and introduce an automated system capable of delivering quicker, more reliable, and precise outcomes.

An annotated panoramic X-ray image was used to train the YOLOv8 model on dental conditions (cavities, fillings, impacted teeth, and implants). The model acquires spatial features like edges, contours, and structural variations that occur within the areas of the teeth so that it is able to differentiate various forms of anomaly. Bounding box annotations can be used to both classify the condition and find its very location in the image, which is critical in clinical interpretation.

Common object detection measures, especially mean Average Precision (mAP), are used to assess the performance of the model. This model had a mAP at 0.5 of 88.7% and a mAP at 0.5:0.95 of 66.3, showing that the model has high detection rates even with varying intersection over union thresholds. The high mAP@0.5 value indicates that the model can correctly identify objects in case there is a reasonable overlap between the predicted and ground truth bounding boxes and the mAP@0.5:0.95 value indicates the performance of the model in more demanding evaluation conditions. These findings affirm that the model can do a good job at detection even in strenuous conditions.

The validation accuracy curves and the training curves give an insight into the way the model learns with time. Figure 4.1.1 depicts that the training accuracy and validation mAP exhibit a continuous upward trend when there is an increase in the number of epochs. This shows that the model is increasingly learning meaningful features and

enhancing its ability to detect. The fact that the training and validation curve are very close to each other indicates that the model will generalize to unseen data, and is not severely overfitted. The minimal differences in the curves are natural and suggest that the model adapts to other samples in the process of training.

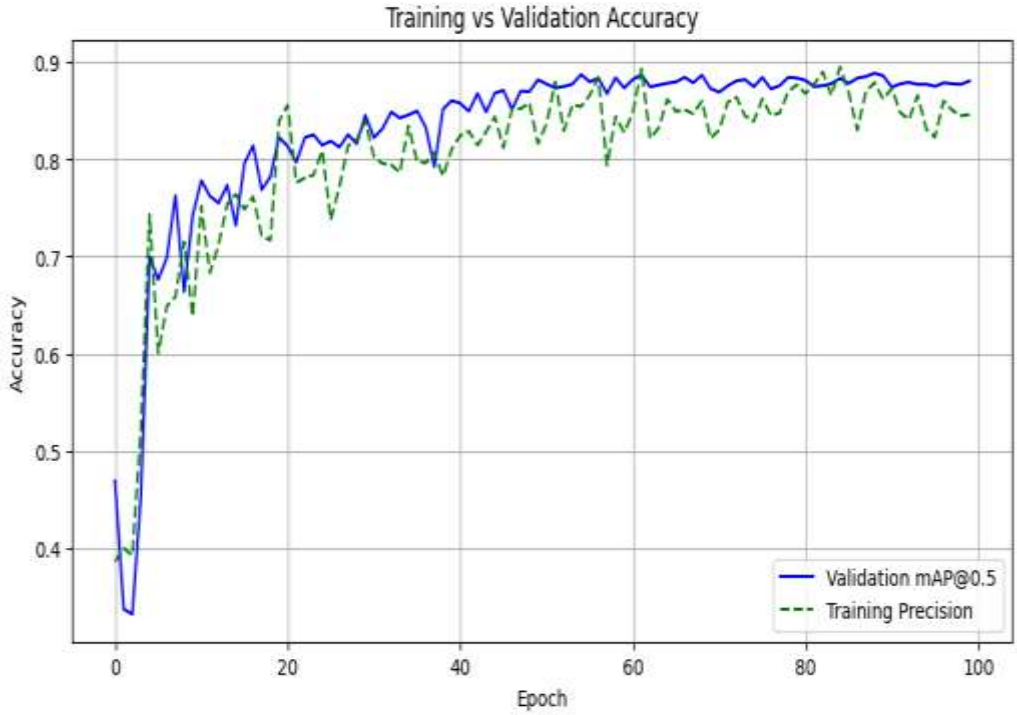

Figure 4.1.1: Training vs Validation Accuracy Curve of YOLOv8 Model.

The behavior of the model with respect to learning is further explained in the loss curves. Figure 4.1.2 illustrates the change in the training and validation loss with the epochs. Box loss and classification loss are steadily reduced, which means that the model is successfully reducing losses on localization and classification processes. The gradual reduction in the values of the losses demonstrates the optimization and effective learning stability. There are no sharp spikes or irregularities in the loss curves, which indicates that the training is well-balanced, and it is not unstable.

The other significant finding of the results is consistency of training and validation performance. Overfitting can be a common problem in numerous deep learning models, where the model fits the training data but fails to fit the validation data. Nevertheless, here, the near coincidence between training and validation curves suggests that the model has similar performance on various datasets. This is a crucial consideration to

the real-world application, as it will guarantee the model can work effectively on invisible images.

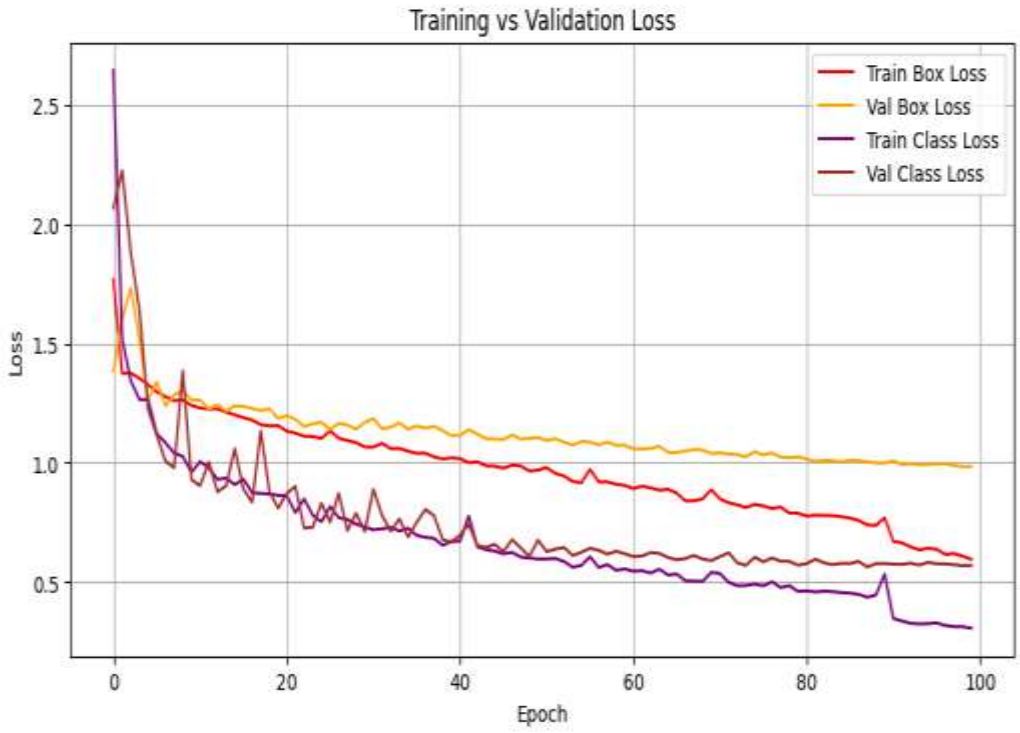

Figure 4.1.2: Training vs Validation Loss Curve of YOLOv8 Model.

The model is also able to differentiate a number of dental conditions well. As dental abnormalities may resemble each other in visual appearance, to fully classify them correctly the model must learn to differentiate minute variations in texture, shape, and structure. The performance attained shows that the YOLOv8 model can capture such fine-grained features. Also, the multi-scale detection mechanism is used to enable the model to recognize objects with different sizes, which is especially helpful in dental pictures, where anomalies can be observed at different scales.

Another significant benefit that is realized in this step is the computational efficiency of the YOLOv8 architecture. Although the model has high accuracy, it has a high inference speed and needs comparatively fewer computational resources than conventional two-stage detectors. This renders it to be appropriate in real time applications where fast diagnosis is needed. The streamlined structure of YOLOv8 will make the system efficient enough to handle images without having to lower its performance.

The final model performance is summarized in Table 4.1 that gives a brief depiction of the main evaluation parameters. The table indicates the effectiveness of the YOLOv8 model in detection accuracy, precision, recall, and the overall performance.

Table 4.1: YOLOv8 Mini-1 Detection Results

Model	Yolov8m
Accuracy	88.7%
Precision	0.86
Recall	0.86
F1-score	Balanced

The findings have made it evident that the detection system based on the YOLOv8 can be used to reliably and precisely find dental abnormalities. The model exhibits high learning, consistent training and excellent generalization of learning. It is computationally efficient and highly accurate, thus appropriate to be used in real-world dental healthcare settings.

In general, this step manages to define the viability of object detection based on deep learning to use automated diagnosis of dental problems. It forms a robust base upon which it can be further improved in the next phases through the addition of other functions like processing of real-life images, tracking of disease progression as well as knowledge-based recommendation systems to enhance the overall functionality of the system.

4.1.2 YOLOv8n-Based Real-Time Dental Detection Results

The second phase of the proposed system aims at expanding the dental disease detection ability of the structured panoramic X-ray images to the real-world intra-oral images with the help of the YOLOv8n model. This step is essential in determining the practical feasibility of the system in practice since real-life dental images usually possess changes in the lighting conditions, image quality, orientation and noise in the backgrounds. The main aim of this phase was to come up with a powerful detection

mechanism that can deal with such variations without compromising its accuracy and reliability. Moreover, the stage adds the disease progression tracking that expands the capacity of the system to trace the changes in the dental conditions in the course of time.

As a lightweight implementation of the YOLO architecture, the YOLOv8n model is a special design that aims at delivering a trade-off between computational efficiency and detection accuracy. Although the model has less parameters than larger counterparts, it performs well in identifying numerous dental conditions at the same time. The training process has stable convergence which means that the model is successful in learning spatial and contextual features of the dataset. The fact that the model is able to operate effectively with limited resources renders it applicable to real-time use, such as web-based healthcare systems.

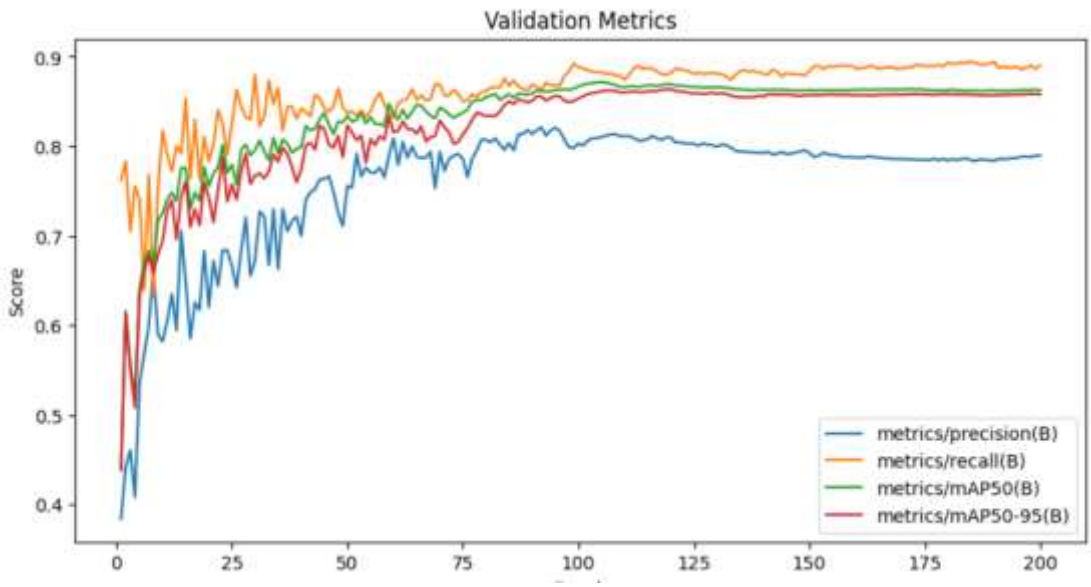

Figure 4.1.3: Accuracy Curve (Precision, Recall, mAP) of YOLOv8n Model.

Validation metrics curve in Figure 4.1.3 gives a complete picture of the model performance in terms of training. The values of the precision, recall, and average mean precision (mAP) demonstrate a steady increase with increasing the number of epochs. The recall values are always high, above 0.90, which demonstrates that the model manages to recognize most of the real dental conditions, which exist in the images. High recall is also required especially in medical application because failure to identify a condition may result in false diagnosis or delayed treatment. The mAP at 0.5

converges in the range between about 0.85 and 0.88, which proves that the model attains a high degree of detection and localization of dental anomalies is accurate. The values of precision also increase continuously, meaning that the number of false positive detections decreases as the model learns more accurate feature representations.

The loss curves in Figure 4.1.4 give more insight into the process of optimization of the model. The loss on training box and classification loss also reduces gradually with each epoch, which means that the model is indeed reducing the errors in both localization and classification. The validation loss also shows a similar pattern and this proves that the model is reliable in generalizing to unseen data. The fact that the loss curves do not have any sudden spikes or irregular curves is an indication that the training process is stable and without problems like gradient instabilities or inappropriate learning rates. The merging of both the training and the validation losses shows that the model is in an optimal condition and further training will lead to insignificant changes.

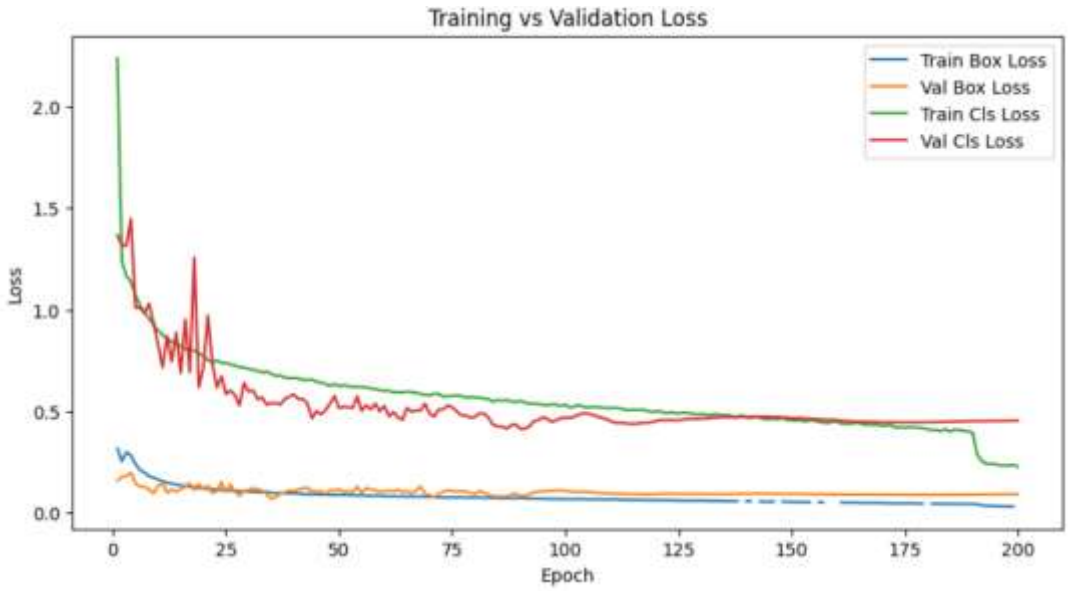

Figure 4.1.4: Training vs Loss Curve of YOLOv8n Model.

Figure 4.1.5, the confusion matrix, gives a breakdown of the performance of the model on each of the classes. The high level of diagonal dominance as seen in the matrix represents that the model effectively categorizes most of the dental conditions. The high true positive rates of classes like caries, calculus, gingivitis, tooth discoloration, mouth ulcer and hypodontia indicate that the model has the capability to differentiate between the various conditions. There are also minor misclassifications between visually similar

classes, as is to be expected in medical image analysis as features overlap. These misclassifications are however minimal and it does not affect the overall performance of the model significantly. The confusion matrix indicates how the model can learn discriminative features and provide accurate predictions in a variety of classes.

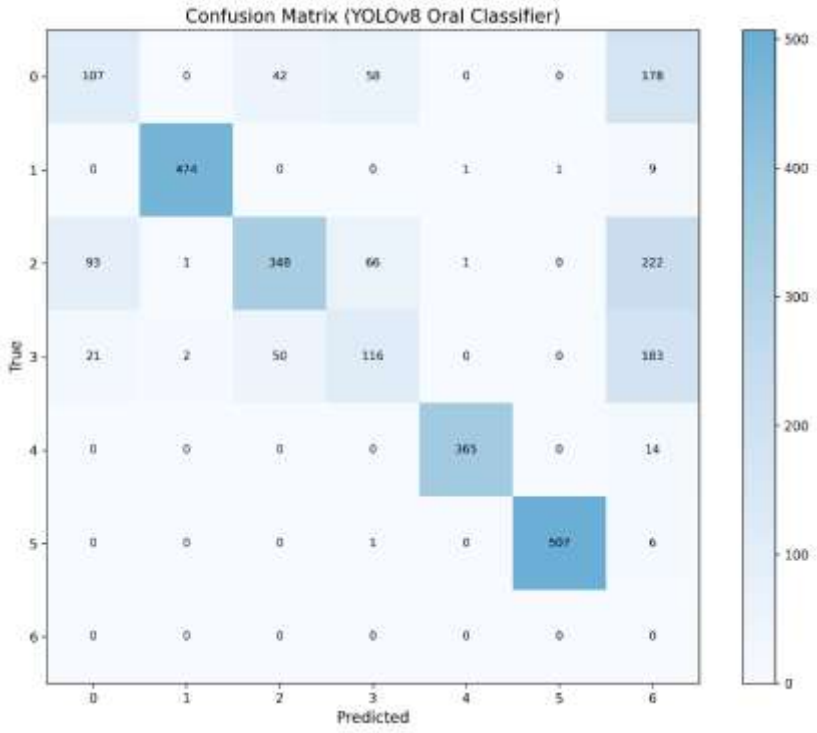

Figure 4.1.5: Confusion Matrix of YOLOv8n Dental Detection Model.

The other important feature of this phase is that the model is very strong to deal with real-life image variations. Contrary to controlled datasets, real-world images can be noisy, occluded, and have varying illumination levels. The overall stability of the performance in the measures of validation suggests that the model has acquired some constant features that remain unchanged by these distinctions. Such strength is critical to its real-world implementation since it means that the system will be able to work under varying circumstances.

Besides detection, this phase presents a disease progression monitoring system, a valuable dimension to the system. The system can examine changes in the severity of dental conditions by comparing bounding box area and the detection confidence of the uploaded images, over multiple image uploads. A rise in the size or confidence score of the detected region spells worsening of the condition whereas a decline implies an

improvement. This feature allows constant control over the health of patients and facilitates early intervention. It also makes the system usable since it gives the insights on how a condition changes with time.

The fact that progression tracking can also be integrated reveals the flexibility of the model outputs to be analyzed by higher levels. The system does not take results of detection as unrelated outputs, but uses them to come up with valuable conclusions about patient health. This transition to the dynamic monitoring of the system instead of merely detecting it is a big leap in the capability of the system.

The overall performance of the YOLOv8n model is summarized in Table 4.2. The key evaluation metrics that are presented in the table, such as mAP, precision, recall, and F1-score, represent the overall efficacy of the model. The recalled values are high hence confirming that the model is able to identify most of the dental conditions, the balance in the precision will, however, ensure that false positives are minimized. F1-score represents the balance of precision and recall which means that the performance of the F1-score is similar across the classes.

Table 4.2: YOLOv8n Determinations.

Model	Yolov8m
mAP@0.5	90%
Precision	85
Recall	90
F1-score	Balanced
Stability	High

In general, the dental image analysis based on the YOLOv8n system has a high performance in real-world. The model exhibits consistent training, effective convergence, and low overfitting. The large recall values guarantee a thorough detection of dental conditions whereas the mAP values guarantee correct localization and classification. The confusion matrix shows consistent performance in terms of

classes, and the progression tracking system provides a lot of practical value to the system.

This step is a significant advancement compared to the first model as it allows actual implementation in the real world and dynamic monitoring. The findings validate the fact that the system itself is scalable and robust and accurate. These developments provide a good background to the last phase in the project where hybrid modeling and knowledge-based recommendation system is incorporated to enhance the intelligence and usability of the system.

4.1.3 Hybrid YOLO–EfficientNet with RAG-Based System Results

The last phase of the suggested system will be the full adoption of the detection, classification, and intelligent recommendation generation into a unified dental healthcare system. The phase integrates the advantages of YOLO-based object detection to localize objects spatially, EfficientNet to classify and analyze the features richly, and a Retrieval-Augmented Generation (RAG) module to make personalized and contextual suggestions. The main purpose of this step was to improve the overall diagnostic capacity of the system by shifting towards more than mere detection and allowing a more comprehensive grasp of the dental conditions and actionable information. Such integration will turn the system into a complete clinical decision-support system that will be able to support patients and practitioners alike.

The EfficientNet model applied in this step shows that it is both a good learner and a good classifier of various dental conditions. The overall accuracy of the model is 94, which means that it can be used to reliably differentiate between the various classes including calculus, dental caries, gingivitis, mouth ulcers, tooth discoloration and hypodontia. The high value of the accuracy indicates that the model is effective to extract discriminative features of intra-oral images such as texture patterns, color variations, and structural differences. This is especially significant in dental image analysis where minor differences between conditions can make a big difference in diagnosis.

The curves of training and validation accuracy (Figure 4.6) give information about the learning behavior of the model. The training accuracy rises very fast in the first epochs

and levels off to 96, whereas the validation accuracy is near 94-95 throughout. The minimal distance between the two curves means that the model is well-generalized to unknown data and is not overfitted. The minimal variations seen on the validation curve are plausible given the differences in validation samples, yet they are within the range of stable values which proves the strength of the model. This behavior shows that the model has attained an optimal balance between the learning capacity and generalization.

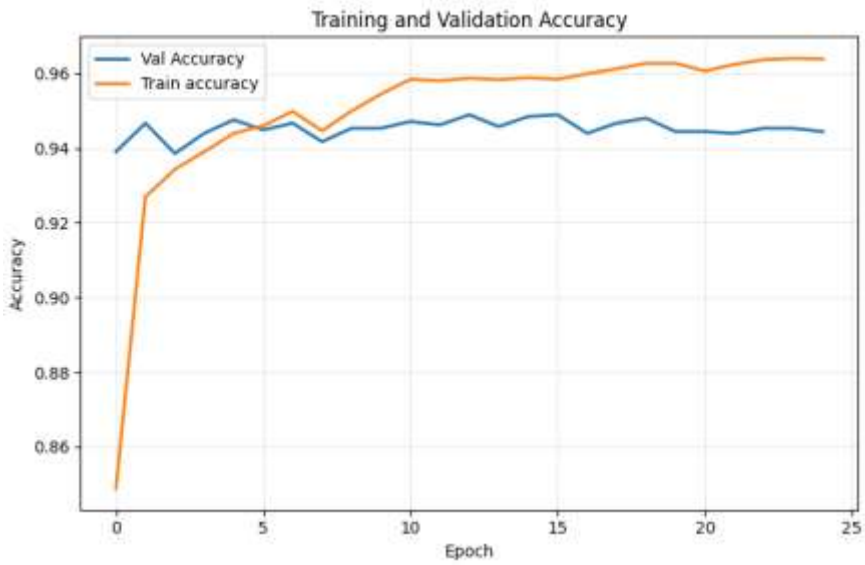

Figure 4.6: EfficientNet Model Training and validity Accuracy Curve.

The effectiveness of the training process can also be substantiated with the loss curves presented in Figure 4.7. The loss in training declines rapidly in the early epochs which implies that basic features are learned fast. The loss steadily reduces and plateaus as training advances, indicating parameter fine-tuning of the model. The loss of validation takes a similar direction, in the sense that the model has the same performance on unseen data. The fact that there is no divergence between training and validation loss shows that the model is not overfitting and the training process is well-regularized. The efficient merging of the loss curves shows that the optimization process was efficient and that EfficientNet architecture was appropriate to this task.

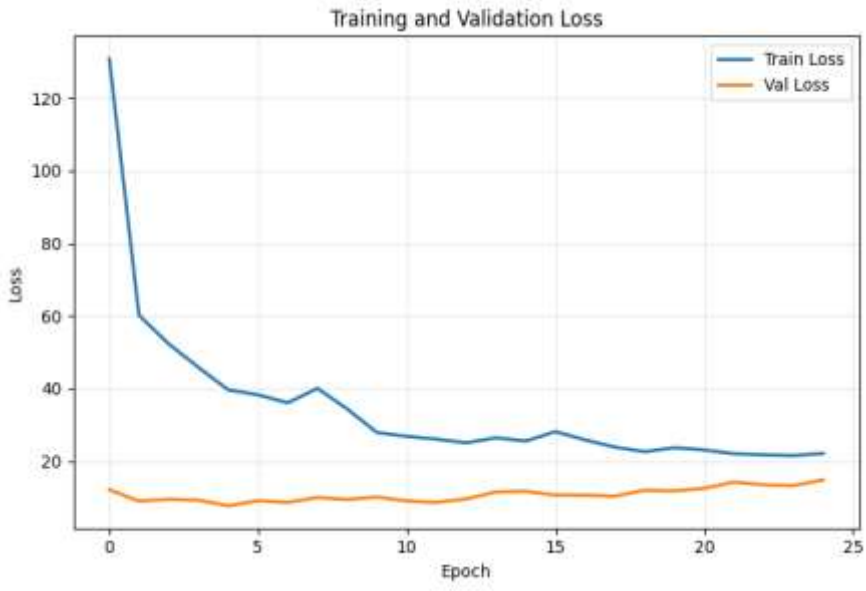

Figure 4.7 EfficientNet Model Training and Validation Loss Curve.

Figure 4.1.8 below shows the confusion matrix that gives a more in-depth evaluation of the model in terms of a class-by-class performance. The high diagonal dominance in the matrix shows that the model is successfully classifying most of the samples in each of the classes. Classes like mouth ulcer, tooth discoloring and hypodontia have near perfect classification and this implies that such conditions have a different visual appearance that can be well captured with the model. Conversely, there is slight confusion in classes like calculus and gingivitis, which are similar in visual characteristics like inflammation and accumulation of plaque. Nonetheless, the rate of misclassification is minimal, and the general performance is very reliable.

The classification report gives additional quantitative evidence of the performance of the model. The values of the precision, recall and the F1-score of most of the classes are approximately or equal to 1.00 which denotes high classification ability. The fact that the macro and weighted average scores of the model are 0.94 proves that the model is constant as well as not biased to any specific category. The marginally less scores in the classes such as calculus and gingivitis can be explained by the fact that the visual patterns are similar and that is a challenge to human experts as well. Nevertheless, the model performs quite well in these instances showing that it is able to acquire complex features.

Figure 4.8: Confusion Matrix of EfficientNet Classification Model.

The results of the detailed classification can be summarized in Table 4.3 that can briefly demonstrate the performance of the model in various classes.

Table 4.3: EfficientNet Classification Results

Model	Yolov8m
Accuracy	94%
Precision	0.95
Recall	0.94
F1-score	0.94
Weighted Avg	0.94

The new feature of this phase is that it combines detection and classification results to analyze the severity. The YOLO model is used to detect the position of dental conditions and the extracted regions of interest are analyzed by EfficientNet to identify the severity level. This combination enables the system to give more detailed and

clinically meaningful insights. The system does not just determine whether a condition exists; it can also determine the severity of the condition, which is essential in planning and making decisions on treatment.

The other major input of this phase is the introduction of the Retrieval-Augmented Generation (RAG) module. This module complements the system by providing customized suggestions, referring to identified conditions and levels of severity. The RAG framework uses a similarity search to find information on the dental knowledge base related to the data input and uses model predictions to produce context-dependent outputs. Such recommendations consist of treatment recommendations, preventive strategies, and daily care routines based on individual conditions. This aspect enhances the system drastically as it is much easier to use because it does not offer detection outcomes but actionable information.

Disease progression is also enhanced through the system by comparing the current and past results which are stored in the database. The system can identify the improvements, worsening, and stability of a condition by assessing the change in bounding box area, classification output, and confidence scores. This feature allows tracking patient health continuously and prevents early intervention. Progression tracking combined with a classification and a recommendation generation system make the system dynamic and intelligent.

Computationally, the hybrid structure proves to be efficient despite being complicated. YOLO model is used to guarantee rapid detection whereas EfficientNet is used to guarantee precise classification and optimized parameters. These models can also be integrated without incurring much computational overhead, hence the system can be used in real-time applications. Optimized architectures make sure that the system can be instantiated on resource-constrained environments without performance loss.

In general, the hybrid YOLO-EfficientNet + the system based on the RAG shows a better performance than the earlier stages. The effectiveness of the proposed framework can be noted by the high classification accuracy, the stability during the training process, the high generalization ability, and the incorporation of more sophisticated functions that include severity analysis, progression track, and customized recommendation generation. It is the final phase of the project, which involves a

combination of various methods to produce a powerful, scalable, and intelligent dental health system that can be deployed in real-life settings.

4.2 Test case results/Result Analysis

This sub-section explains the way the developed dental disease detection system behaves in real-time when real input images are given. Whereas the earlier section centered on quantitative assessment measures like accuracy, loss curves and confusion matrices, the present section concentrates on the actual functioning of the system. It outlines the manner in which the model works with actual input images and produces meaningful outputs.

This section aims at filling this gap between theory and practice by showing how the system would convert raw dental images into usable diagnostic data. It emphasizes the relations between input, model processing, and the output generation, thus, confirming the usefulness of the proposed framework in the real-world application.

4.2.1 YOLOv8-Dental Detection System Result

The results of the dental detection system based on YOLOv8 using a panoramic dental X-ray image as input are depicted in Figure 4.9. This step of the system aims to detect and localize dental diseases (cavities, fillings, and implants) with a deep learning-based object detector model. The system then processes the image and marks the identified regions with bounding boxes and the respective class labels and the confidence scores. Such visualized results can aid in comprehending the way this model perceives the input image and detects various dental abnormalities.

Figure 4.9: YOLOv8-Detection Result with Bounding Boxes.

The test case starts with the input of a panoramic image of dental X-ray as seen in Figure 4.10. This image is the raw and unprocessed data that is given to the system. It consists of the complete dental structure with alignment of the teeth, jaw construction, and any deviation of teeth like cavity or implants. The input image plays a crucial role in the analysis because it enables the reader to learn about the intricacy of the data that the model is working on. Dental X-ray images are usually characterized by overlapping structures and different levels of intensity and therefore detecting them is not an easy task.

Figure 4.10: Input Panoramic Dental X-ray Image.

After providing an image, it is subjected to preprocessing, i.e. resizing, normalization, and formatting to suit input requirements of the YOLOv8 model. All these preprocessing operations take care of the standardization of the image and its readiness to be efficiently processed by the neural network. The model is then used to extract features after preprocessing which are spatial patterns including edges, contours and intensity variations. The features play a vital role in diagnosing dental conditions because various abnormalities have different visual appearances.

The YOLOv8 model uses only a single forward pass to process the entire image, therefore it is able to identify multiple objects at the same time. It breaks down the image into parts and estimates bounding boxes and class probabilities of every part. The model subsequently uses confidence thresholds and non-maximum suppression to filter out redundant detections and only the most pertinent predictions. This will give a set of final detections that will be a good representation of the dental conditions in the image.

Figure 4.11 shows the output of the system, with the identified dental conditions highlighted with bounding boxes. All of the bounding boxes contain the predicted class (cavity, fillings, or implant) and a confidence score which reflects the trustworthiness of the prediction. The fact that there were several bounding boxes on the output shows that the model can identify several conditions in a single image. This is especially so in the analysis of the dentals where various anomalies can exist in varying parts of the mouth.

Besides a visual output, the system also provides a formatted report of identified conditions. All the dental anomalies identified have been summarized in a clear and concise way and this makes it easy to interpret the results by the users. As an example, the system can recognize a condition like cavities, fillings, and implants and display them in a tabular format. This enhances the usability of the system, as users can quickly understand the detected conditions without analyzing the image manually.

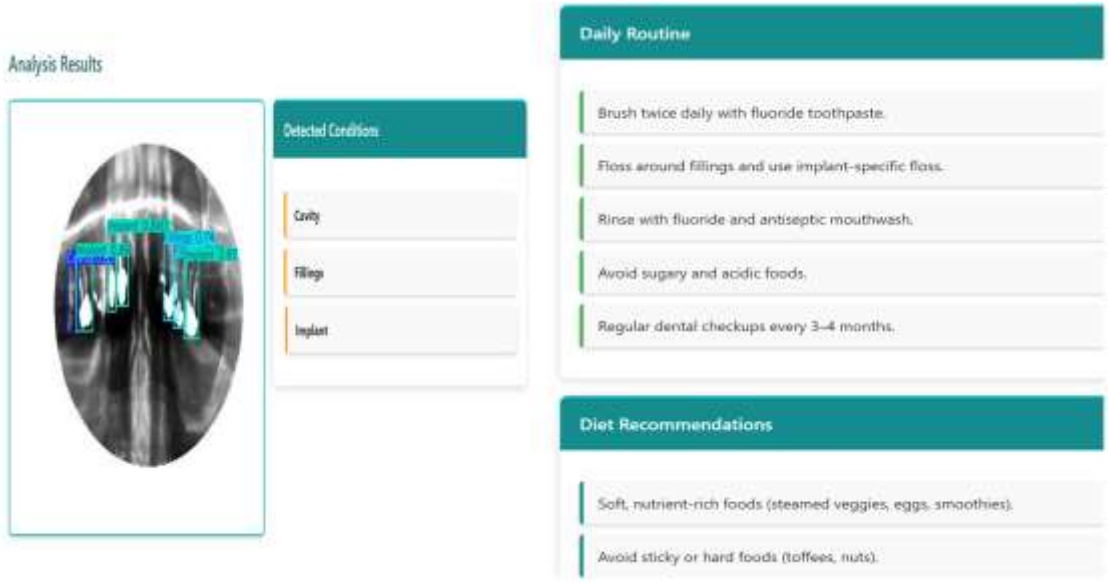

Figure 4.11: Dental Conditions and Output Summary detected.

This test case analysis shows that the system can effectively identify and localise dental conditions in complex panoramic images. The model manages to detect the various types of anomalies and separates the anomalies using learned features. The confidence scores that are attached to each detection give an added degree of reliability that enables users to determine the accuracy of the predictions. When the confidence values are high, it means that they are good predictions and when they are low, they might need additional confirmation.

The other thing that can be remembered in this test case is that the model can cope with the differences in the quality and structure of the images. Dental X-ray images are characterized by noise, different contrasts, and overlapping of structures. Despite these challenges, the model is able to extract meaningful features and produce accurate detections. This shows that the model has acquired strong representations that are not highly influenced by changes in input data.

The system is also efficient in processing the input image. As a single-stage detector, YOLOv8 also does not require multi- passes to detect and classify, which makes it faster to infer than the traditional multi-stage models. This makes the system applicable in real-time applications where expedient analysis is needed. Speed and accuracy are the major benefits of the YOLOv8 architecture.

Practically speaking, the findings of this test case point to the possibility of the system being used as an automated diagnostic instrument. Quick and accurate detection of dental conditions would help dentists to detect problems at an early stage. It may also be applied as an aid to patients, giving them initial information prior to seeking professional help. The system lessens the reliance on manual inspection and enhances consistency in the diagnosis.

Nevertheless, some limitations that are presented in testing should also be taken into consideration. The model can also yield overlapping bounding boxes or slightly inaccurate localization in certain scenarios because there are similarities in visual features. Also, subtly different conditions might need additional fine-tuning of the model to increase the accuracy of classification. These drawbacks denote potential improvements that can be made in the future including diversifying datasets, and extending feature extraction methods.

In general, the outcomes of the test case confirm the usefulness of the dental detecting system based on YOLOv8. The system has been able to convert a raw dental X-ray image into meaningful diagnostic information by identifying and labeling dental anomalies. The system is very appropriate to be deployed in practice due to the combination of the proper detection, efficient processing, and useful output. This phase provides a robust base to build on successive improvements at the later phases where other functionalities like real world image processing, disease progression monitoring, and generation of personalized recommendation are added to establish a full dental healthcare system.

4.2.2 YOLOv8n-Based Real-Time Detection and Recommendation Results

Figure 4.12 presents the results of the dental detection system implemented with YOLOv8n as the dental detector in case a real-world intra-oral image is used as the input. This step involves the creation of the system to process images that have been taken in real-life conditions and identify dental anomalies that include calculus, caries, gingivitis, tooth discoloration, ulcers, and hypodontia. This stage, as compared to the

previous stage that was based on structured panoramic X-ray images, is based on images that possess natural variations in lighting, angle and texture. This complicates the detection task to be more realistic and difficult which justifies the strength and feasible applicability of the model.

Figure 4.12: YOLOv8n Detection Result on Intra-Oral Image.

The test case starts by providing an intra-oral dental image which is displayed in Figure 4.13. This image is a raw data that is taken in a real-world situation and given to the system by the user. There are clear indicators of dental diseases in the image including plaque formation, tooth discoloration and possible gum diseases. The presence of such variations makes it difficult to manually identify conditions, thereby highlighting the need for an automated system. The input image is used as the reference point of measuring the effectiveness of the model when processing complex visual data.

The input image is preprocessed before being processed by the model; the preprocessing steps include resizing, normalization and noise handling. These operations guarantee the image to be compatible with the YOLOv8n model and enhance the quality of features extraction. The model further examines the photograph to identify spatial characteristics of the photograph like edges, gradients, color

intensities and texture patterns. These characteristics are critical in the process of diagnosing various dental diseases since each disease has its own visual features.

Figure 4.13: User-provided input Intra-Oral Dental Image.

YOLOv8n model follows the method of detection by cutting the picture into grids and forecasting bounding boxes and the probability of the class of each region. It then uses confidence thresholding and non-maximum suppression to eliminate redundant detections and only keeps the most accurate predictions. The result displayed in Figure 4.2.4 indicates the identified area which represents calculus, and a confidence value of about 0.56. The confidence score is moderate but this is still a valid detection using the learned features.

Among the strengths that are evident in this stage, there is the fact that the model is able to identify conditions even when they are not very salient. In real life images, dentals anomalies might not necessarily be noticeable in the image as a result of lighting or occlusions. Nevertheless, the model has been able to find the relevant regions despite these challenges, and hence it is robust and has the ability to generalize.

After the detection process, the system will produce a well-organized and easy to use output interface as illustrated in Figure 4.14. This interface shows the diagnosed condition and a customized dental care routine. The recommendations are specific to the identified condition and are defined in a time-based manner, such as morning,

afternoon, and night routines. Such a systematic method improves the system usability as it offers understandable and practical guidance.

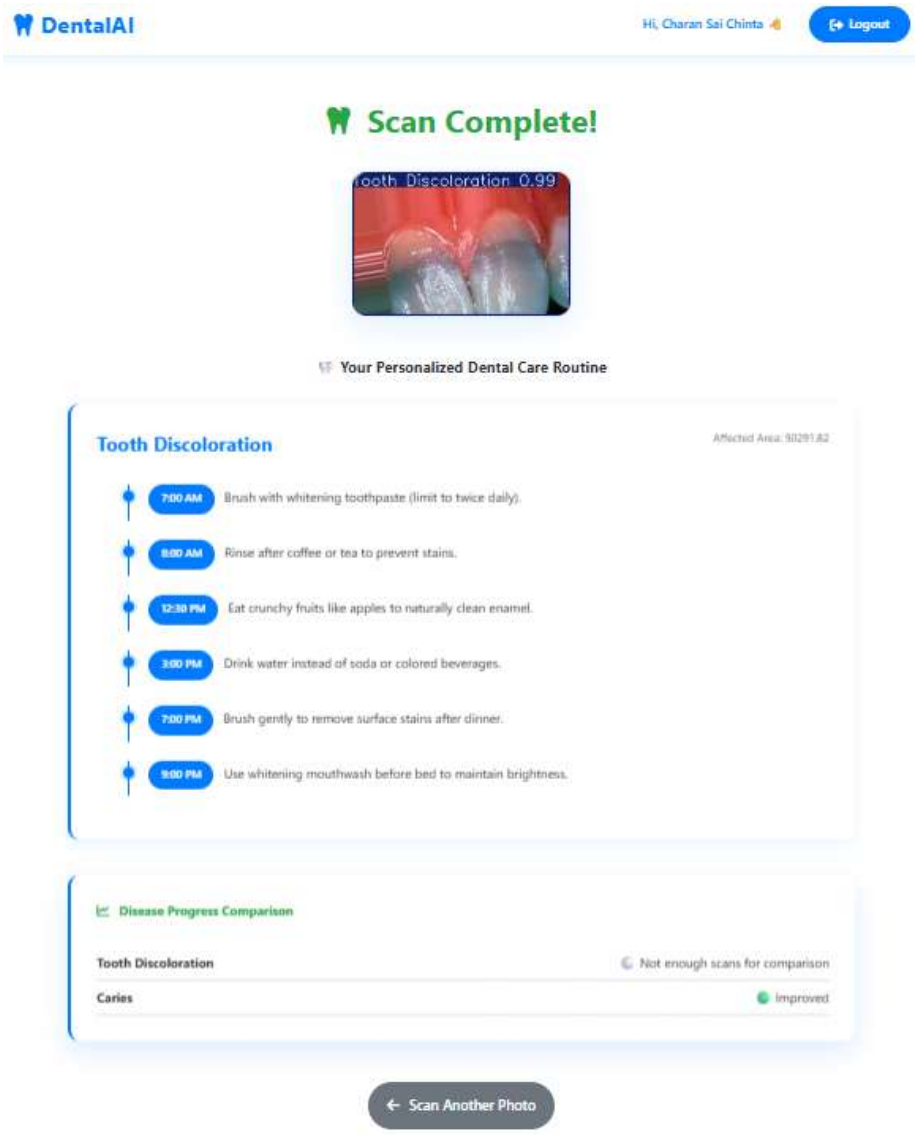

Figure 4.14: Personalized Daily Care Recommendations with Detected Condition.

The guidelines consist of viable recommendations, like brushing, fluoride toothpaste, post-meal rinses, avoidance of specific foods, and oral hygiene. Time-based scheduling aspect makes sure that the user can easily integrate such recommendations into his/her daily life. This capability greatly contributes to the practical usefulness of the system, as it transcends detection and offers actionable healthcare guidance.

The capability of the system to create recommendations of multiple dental conditions at a time is another crucial capability that has been demonstrated in this stage. The

system offers various conditions like calculus, caries, gingivitis, mouth ulcers, tooth discoloration, and hypodontia separate recommendation modules as shown in Figure 4.15. Conditions have a specific set of recommendations, making sure the user is provided with a full-fledged guidance.

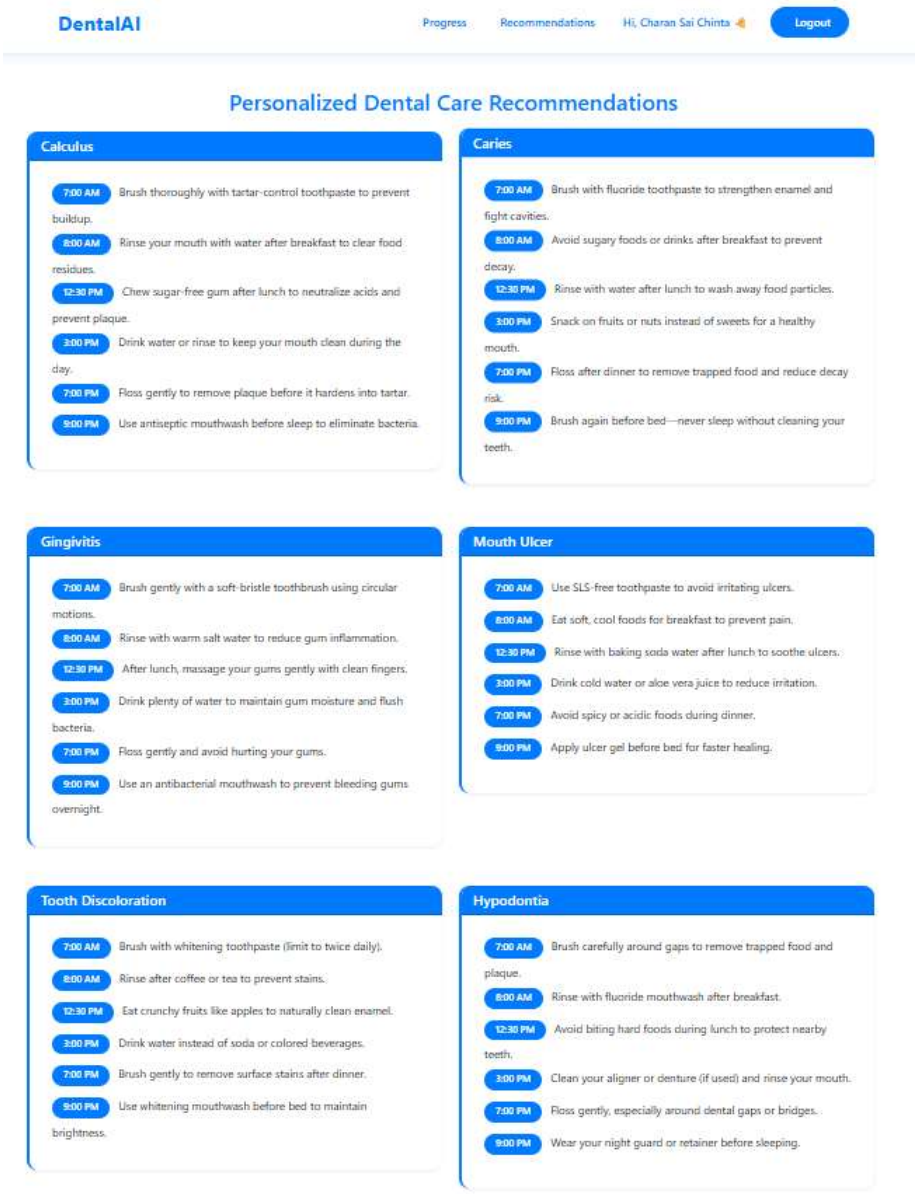

Figure 4.15: Recommendations of individual cases of various dental conditions.

Such a multi-condition recommendation feature is especially useful in practice, when a patient can have a combination of more than one dental problem simultaneously. The system is efficient in sorting out recommendations on each condition in such a way that

there is no confusion or ambiguity. Such formatted output enhances the user experience and makes the system more practical to use on a daily basis.

Besides detection and recommendation, the system has disease progression tracking. The system can identify whether a condition is improving, worsening, or stable by comparing the present results on detection to the previous results that are stored in the database. This is an important feature to enable people to keep track of their health and to see how effective their treatment or care regimen is in the long term.

Technically, the YOLOv8n model has been shown to be efficient in terms of both speed and accuracy. It is a lightweight model, which uses fewer computational resources yet has a high detection performance. This renders it usable in web based applications and real time systems. High speed image processing guarantees users immediate feedback that is needed in real life healthcare applications.

The other important finding during this test case is that the model can generalize on the various types of images. The steady output under different lighting conditions and image qualities depict that the model has acquired strong feature representations. This is an essential need to actual deployment because the system should be able to accommodate various inputs by various users.

Nonetheless, some shortcomings are also witnessed during this stage. Some detections can have lower confidence scores, because of ambiguity in visual features, or a lack of training data on particular conditions. Also, there can be minor errors in the place where the bounding boxes are located, in situations where the boundaries of the condition are not well defined. These weaknesses indicate that further advances are necessary, including diversifying the datasets and training models more effectively.

In spite of the limitations, the system performance is very satisfactory in general. The detection, recommendation, and progression tracking can prove that the system is capable of operating as a complete dental healthcare solution. The model is able to transform raw input images into meaningful outputs that contain diagnostic information and actionable advice.

All in all, the outcomes of the test case confirm the efficiency of the system based on YOLOv8n in practical situations. The system shows a high level of detection, high performance in different conditions and increased usability with personalized recommendation. This phase is a major improvement on the first model as it integrates the technical with the practical applicability, thus leading to the third and last phase where hybrid modeling and smart recommendation systems are further refined.

4.2.3 Hybrid YOLO–EfficientNet with RAG-Based System Results

Figure 4.16 demonstrates the results of the suggested hybrid dental disease detector system in case of a real intra-oral image is used as an input. It is the last and most developed stage of the system where several components are combined together such as YOLO-based object detection, EfficientNet-based classification, and Retrieval-Augmented Generation (RAG) to give a complete and intelligent dental healthcare solution. This system is multi-level in contrast to other stages in the past which concentrated on detection and did not utilize multi-level analysis since it involves spatial localization, feature-based classification, severity estimation, progression tracking, and personalized recommendation generation.

Figure 4.16: Final Image with identified Dental condition.

The resultant image vividly illustrates how the YOLO model can be used to identify and pinpoint dental anomalies in the intra-oral image. A boxing area with a confidence score is used to indicate the accuracy of the prediction. The system can detect different

dental diseases including calculus, caries, gingivitis, ulcer, discoloration, and hypodontia. Localization of these conditions is important in diagnosis since it enables users to have visual knowledge on the affected area.

Visual representation is not the only way to visualise the detection output. The system itself identifies the region of interest (ROI) that matches every identified bounding box, and feeds it to the EfficientNet model to continue the analysis. This extra processing step makes the system more accurate as it refines the classification results and allows the estimation of severity. Detection and classification make sure that the system is able to give the system spatial and contextual information of dental conditions.

After the visualization of the output, the input image according to the input image is presented in Figure 4.17. This is a photo of the intra-oral dental image that the user originally uploaded prior to the processing of the image. The natural variations in the input image include variations in lighting, shadows, reflections, and variations in texture, which are typical in real life. These variations make the detection task more complex and highlight the importance of robust feature extraction.

Figure 4.17: Intra-Oral Dental Image, input by user.

The input image is processed by the system in various steps. First, preprocessing activities of image resizing, normalization and noise reduction are performed so that the image fits into the model. The YOLO model then extracts features by examining features like edges, gradients and variations in intensity. It splits the image into grids

and predicts bounding boxes and class probabilities of each region. Non-maximum suppression is used to eliminate redundant detections and only the most useful predictions are used.

After the detection step is over, the extracted areas are sent to the EfficientNet model that runs classification and severity analysis. The EfficientNet employs deep convolutional layers to produce high-level features and distinguish between similar dental conditions. This is especially necessary in cases like calculus and gingivitis that might share visual similarities. Integrating the YOLO and EfficientNet, the system has higher accuracy and reliability than when a single model is used.

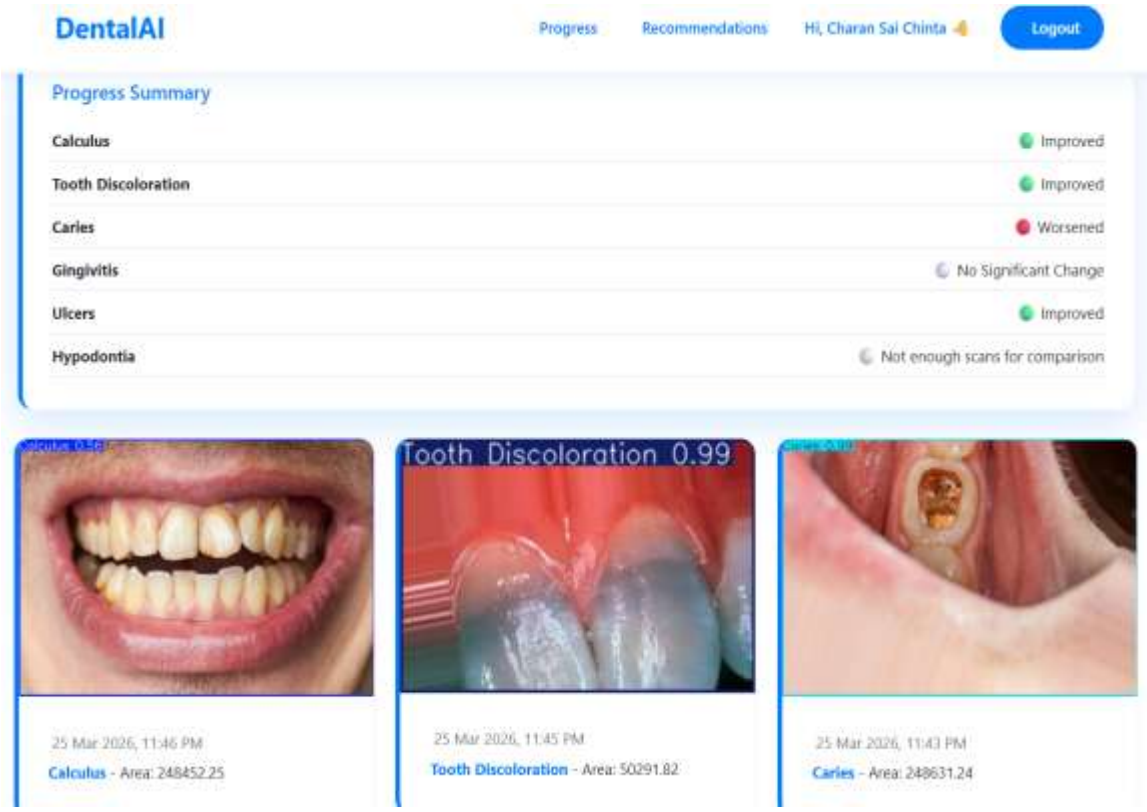

Figure 4.18: Dental Disease Detection and Progress Tracking Dashboard.

Once the detection and classification phase is complete, the system will produce an elaborate dashboard interface as presented in Figure 4.18. This dashboard gives an overview of conditions identified and their progression status. It has various elements like condition list that has been detected, status indicators, and results of comparison. The dashboard is a central interface that coordinates all outputs of the system into a single one.

The dashboard shows different dental diseases like calculus, tooth discoloration, caries, gingivitis, ulcers and hypodontia. A status indicator is provided to each condition to indicate whether the condition has improved, worsened or remained stable. The color-coded markers are used to represent these indicators and make the output easier to read. To illustrate, green means improvement, red means worsening and grey means no change of significance.

The most important contributions of this stage are the progression tracking feature. The system compares the present results of detection with the past history in the database. The system identifies the progression of each condition by examining the change in area of bounding box, confidence scores and the classification results. This enables users to keep a track of their dental health over time and determine the effectiveness of any treatment or lifestyle change.

Besides the dashboard, the system has an elaborate recommendations interface as demonstrated in Figure 4.19. This interface is created with the help of the Retrieval-Augmented Generation (RAG) module, which integrates information retrieval and natural language generation to create personalized recommendations.

The recommendations are specific to each identified condition and they are provided in a systematic format. In all the conditions, the system offers a routine of daily care that involves certain activities like brushing, flossing, rinsing and dietary modifications. The recommendations are categorized in terms of time intervals like morning, afternoon, and night and are simple to follow.

To illustrate, the system prescribes a tooth whitening paste, avoidance of colored drinks, and good oral hygiene in the case of tooth discoloration. In some cases like caries, the guidelines involve minimizing sugar consumption, applying fluoride toothpaste and regular dental check-ups. All the recommendations are aimed at the particular needs of the identified condition, which enhances the effectiveness of the guidance.

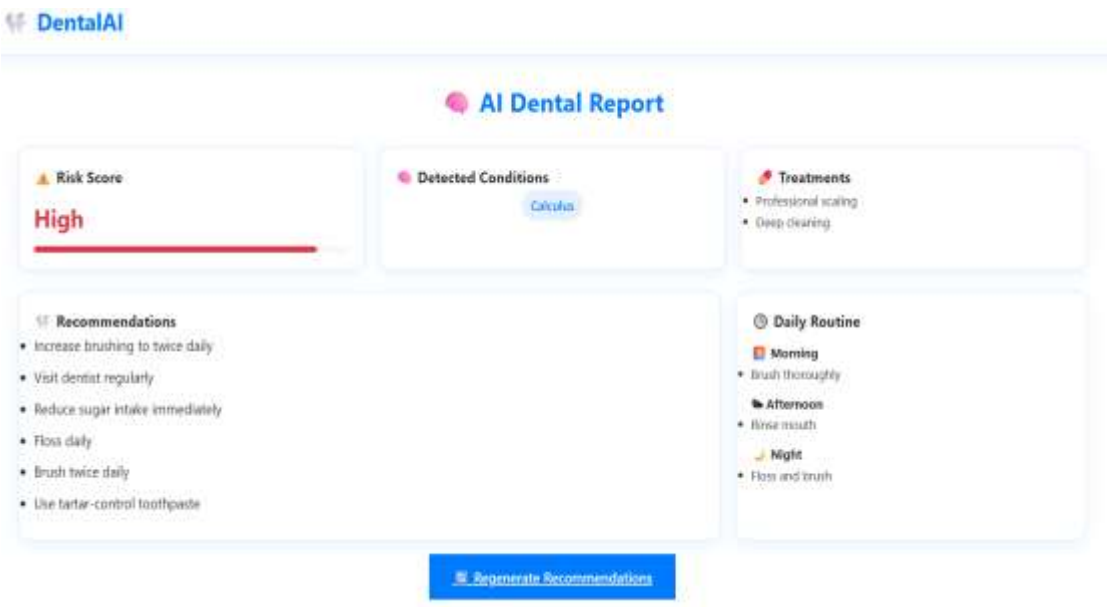

Figure 4.19: Multiple Dental Conditions Interface of Recommendations.

RAG module is vital in generating such recommendations. It finds similarity search models like FAISS to retrieve relevant information in a pre-built dental knowledge base. The retrieved information is then fed through a language generation model to give recommendations in a human readable form. This makes the output accurate and easy to comprehend.

The capability of the system to deal with a number of conditions at a time is another key feature of this stage. In real-world scenarios, patients often have more than one dental issue. The system identifies all the conditions within the image and makes individual recommendations. Such a multi-condition management power guarantees a thorough healthcare provision and eliminates the possibility of missing vital information.

The system has also shown high levels of generalization, whereby it works well on images that are taken at varying conditions. The capability to manage changes in lighting, angle, and image quality denotes that the model has acquired some powerful features. This is necessary in the real world where the conditions of input are beyond control

Computationally, the hybrid system is efficient even though it is very complex. YOLO model provides quick detection, and EfficientNet offers precise classification with

optimal parameterization. The RAG module is efficient as it accesses only pertinent information, which reduces the overhead on computation. This makes sure that the system is able to provide real time results with minimal delays.

The system is user friendly and its user interface is easy to understand. It presents the information in a straightforward and tabular form enabling the users to quickly understand the findings. Visual outputs, textual summaries, and structured recommendations complement each other to make the user experience better. This renders the system applicable in clinical and personal healthcare applications.

Nevertheless, some shortcomings are realized in this phase. Recommendations rely on the quality and completeness of the knowledge base to make accurate recommendations. Moreover, certain conditions where the differences are not very pronounced can lead to a minor decrease in the confidence scores. These shortcomings illustrate the necessity to further improve them by adding more data and elaborating the knowledge base.

Nevertheless, the performance of the whole system remains very satisfactory despite these limitations. The system is a comprehensive solution to dental healthcare as it integrates detection, classification, progression tracking, and recommendation generation. The system manages to turn raw input images into useful outputs that give both the diagnosis and practical advice.

In general, the test case findings substantiate that the suggested hybrid YOLO-EfficientNet with RAG-based system can be used to conduct an end-to-end analysis of dental in real-life applications. The system is very precise, strong and usable and hence is applicable in practice. This phase is the final point of the project, and many sophisticated methods are integrated to develop a smart and scalable dental health solution.

4.3 Observations from the Work

Based on the observations made in the general development of the proposed dental healthcare system, the framework has demonstrated considerable improvements in all respects of technical performance, as well as the practical usability, interpretation of

findings, and applicability in real-life. With each step of the project development, the system has evolved into a simple model of the dental anomaly detecting system, and the entire hybrid framework that could detect, classify, track progress, and generate personal recommendations. The very fact that this gradual development is one of the most significant ones proves that detection or classification cannot be the only effective means of dental healthcare systems. Instead, they need a mix of the correct visual analysis, strong feature extraction, insightful interpretation of findings, and practical healthcare advice.

The first and the most critical discovery of this work is that deep learning in dental disease detection is very effective as compared to the traditional manual method of diagnosis. In stage one, the YOLOv8-based detection system displayed a high level of detection of dental abnormalities including cavities, fillings, and implants based on panoramic X-ray images. The obtained results (mAP at 0.5 of 88.7) prove that the model could learn meaningful spatial patterns and properly localize dental conditions. This finding underscores the possibility of deep learning models to make dental diagnosis automatic and less reliant on manual examination.

The other important observation during the first stage is that there are no equal dental conditions that contribute to the overall detection performance. Distinct structural differences were observed more precisely because of their different shapes and textures; this was the case with conditions like implants and fillings. Conversely, such conditions as cavities can occasionally be problematic with their differences in appearance and intersecting with surrounding structures. This shows that the performance of detection models does not only rely on the architecture but the visual distinguishability of the classes as well.

The training curves analysis during the first stage also offers some insights. The gradual improvement in training and validation accuracy and the gradual reduction of the loss values, demonstrate stable learning behavior. The fact that training and validation curves are closely matched indicates that the model can generalize well and also is not overfitted. This is a significant finding, as it proves that the model can work with invisible data, which is crucial to apply it to the real world.

With the project entering the second phase, when YOLOv8n was being applied to real-world intra-oral images, an important observation was that robustness is essential when it comes to real-world variations. Intra-oral images have variations in light, angle, and texture unlike controlled X-ray images. Nevertheless, the model retained high recall values (more than 0.90) and had an mAP between 0.85 and 0.88. This shows that the model has acquired invariant characteristics that enable it to work in various circumstances.

The other significant conclusion of this phase is that detection is not practical in healthcare applications. Although the model could successfully diagnose dental conditions, it was able to provide more effective results in the situation when the recommendation generation was implemented. The system offered an individualized dental care regimen founded on the identified issues, such as brushing habits, eating recommendations, and precautions. Herein lies a key point: the users need not just a diagnosis but practical advice.

The second stage marked a significant observation as well with the introduction of disease progression tracking. The system would then be able to determine if a condition was improving, worsening or stable through the results of multiple scans. This characteristic gives the analysis a time aspect making the system not a static diagnostic tool but a dynamic monitoring system. This feature is especially useful in system applications in the medical field, since constant monitoring is crucial.

The other important observation during the second stage is that precision and recall are important to balance. Whereas high recall is used to guarantee that most of the conditions are identified, precision is used to guarantee that the number of false detections are reduced. The model demonstrated a good balance between these metrics, resulting in a stable F1-score. This means that the system can give credible predictions, and at the same time not produce too many false alarms.

When the project reached its last phase, the integration of YOLO, EfficientNet, and RAG modules led to considerable improvements in performance and functionality. Among the most significant ones, the fact that hybrid architectures are more effective in comparison with single-model approaches was noted. YOLO used in detection and EfficientNet used in classification enabled the system to record a total accuracy of 94%.

This improvement demonstrates that combining spatial localization with deep feature extraction enhances the overall performance.

The other significant learning of the hybrid approach is that both the local and the global feature representations are important in the analysis of the dental image. YOLO obtains both spatial and structural data, whereas EfficientNet retrieves high-level features that allow distinguishing among similar conditions. The combination enables the system to have higher classification accuracy particularly with visually similar conditions.

Another important observation is brought up by the integration of the RAG module: the relevance of contextual intelligence within healthcare systems. Not only does the system identify and categorize conditions, but also recalls pertinent medical information and provides personalized advice. This makes the system to be a decision-support tool and not merely a prediction model. It draws attention to the fact that practical healthcare setting demands the analytical and interpretive abilities.

The other important observation is connected with the correlation between the performance metrics and practical usefulness. Although the classification models can give high accuracy values, the object detection models with recommendation systems are more practical. It means that a greater numerical performance does not necessarily lead to a greater applicability in the real-life. Rather, the more important systems are those that have meaningful and practical outputs.

In the last stage, the user interface is also important to give some insights. The system enables the user to post photos, see detected diseases, monitor disease progression, and get individualized recommendations via a user-friendly interface. This shows that the system is technically viable and can be used operationally as well. The combination of the backend models and a frontend interface makes it possible to use the system in the real world.

The other insight is that data diversity and quality are important. The system has its performance affected by the diversity of the images that are used to train the system. Conditions that are better represented and of more heterogeneous data are more accurately detected, whereas less represented conditions can perform a bit worse. This

underscores the fact that greater and more balanced datasets should be used to further enhance the system.

It is also found in the analysis that the confidence thresholds are significant in detection performance. Choosing a proper threshold assists in compromising between the false positives and false negatives. This is specifically crucial in healthcare where misleading forecasts can be very severe.

Application wise, the system has a high potential of real world implementation. It can be used as a support tool for dentists, as well as a self-monitoring tool for patients. Its capability to deliver immediate feedback and individualized recommendations makes it very beneficial in preventive care.

Scalability of the system is another significant observation made. The modular design enables the inclusion of new features that include integration with the IoT devices, mobile applications, and cloud-based data storage. This renders the system flexible to the advanced healthcare technology in the future.

Lastly, the overall work shows that deep learning with domain-specific knowledge can boost the performance of the system significantly. The combination of detection, classification, tracking of progression, and the generation of recommendations is a holistic solution that aims at handling the technical and practical issues of dental healthcare.

CHAPTER – 5

CONCLUSION & FUTURE STUDY

5.1 Conclusion

The proposed dental disease detection and recommendation system presents an intelligent and efficient solution for early diagnosis, continuous monitoring, and personalized dental care. The system is developed in multiple stages, evolving from a basic object detection model to a hybrid architecture integrating deep learning and knowledge-based reasoning. This progression ensures improved accuracy, robustness, and real-world applicability.

In the initial phase, a YOLOv8-based model was used to detect dental anomalies from panoramic X-ray images. The model achieved a high mAP@0.5 of 88.7, demonstrating strong capability in accurately identifying conditions such as cavities, fillings, and implants. This phase established the effectiveness of automated detection systems over traditional manual diagnosis by providing faster and more consistent results.

The second phase extended the system to handle real-world intra-oral images using YOLOv8n. Unlike structured X-ray images, real-world data includes variations in lighting, angle, and texture, making detection more challenging. Despite these variations, the model maintained stable performance with high recall and consistent accuracy, proving its robustness. Additionally, disease progression tracking was introduced, enabling the system to monitor changes in dental conditions over time. This feature enhances the system's practical value by supporting continuous health monitoring and timely intervention.

A key advancement in this phase is the introduction of personalized recommendations. Based on detected conditions, the system provides tailored daily care routines and preventive measures. This transforms the system from a diagnostic tool into a guidance system, helping users take actionable steps to improve their oral health.

In the final phase, a hybrid model combining YOLO-based detection, EfficientNet classification, and a Retrieval-Augmented Generation (RAG) module was developed. The EfficientNet model achieved an accuracy of 94%, improving the system's ability

to classify dental conditions and assess their severity. The integration of detection and classification ensures both spatial and contextual understanding of anomalies.

The RAG module further enhances the system by generating context-aware and personalized recommendations. By retrieving relevant information from a dental knowledge base and combining it with model predictions, the system delivers accurate and clinically meaningful guidance tailored to individual users.

One of the major strengths of the system is its ability to detect and manage multiple dental conditions simultaneously. This reflects real-world scenarios where patients often experience more than one issue at a time. The system also demonstrates strong generalization capability, maintaining stable performance across diverse datasets and input conditions.

From a usability perspective, the system features an integrated and user-friendly interface that allows users to upload images, view detected conditions, track disease progression, and follow personalized care plans. This makes the system accessible to both healthcare professionals and general users.

Technically, the use of optimized architectures such as YOLOv8 and EfficientNet ensures a balance between accuracy and computational efficiency, enabling real-time performance and practical deployment.

However, certain limitations exist. Model performance depends on dataset quality and diversity, and conditions with limited training data may show reduced accuracy. Additionally, the effectiveness of recommendations relies on the quality of the knowledge base used in the RAG module.

Overall, the system successfully achieves its objectives of accurate detection, reliable classification, effective monitoring, and personalized recommendation generation. It demonstrates the potential of artificial intelligence to enhance dental healthcare by making diagnosis and preventive care more accessible and efficient.

In conclusion, this work represents a significant step toward intelligent dental healthcare systems. By combining deep learning with knowledge-based methods, it

provides a comprehensive and practical solution that improves diagnostic accuracy while delivering actionable insights for better oral health management.

5.2 Future Study

The proposed dental disease detection and recommendation system demonstrates strong performance and practical applicability; however, several enhancements can further improve its accuracy, scalability, and real-world impact.

One key area for improvement is dataset expansion and diversification. Increasing the number of training samples and including images from different populations, age groups, and imaging conditions (lighting, resolution, devices) can significantly enhance model generalization and reduce bias, especially for underrepresented dental conditions.

Another important direction is extending the range of detectable diseases. Currently focused on common dental issues, the system can be expanded to include more complex conditions such as periodontal diseases, oral cancer, and enamel erosion, making it more clinically comprehensive and valuable.

The system can also be improved through advanced severity analysis. Instead of basic classification, future work can use segmentation and regression techniques to provide more precise and continuous measurement of disease severity, enabling better clinical interpretation.

Enhancing disease progression tracking using time-series models can allow the system to analyze changes over time more effectively and even predict future disease trends, supporting early intervention.

The recommendation module (RAG) can be further strengthened by integrating updated medical knowledge bases and advanced language models. Incorporating user-specific data such as age, medical history, and lifestyle factors can enable more accurate and personalized recommendations.

Future development may also focus on mobile and real-time applications, allowing users to capture and analyze dental images through smartphones and enabling real-time video-based diagnosis in clinical environments. This will significantly improve accessibility and usability.

Improving model explainability using visualization techniques can increase trust among healthcare professionals by showing how decisions are made. Additionally, optimizing models for edge devices will enable deployment in low-resource settings.

Finally, clinical validation and system integration are essential. Testing the system in real-world clinical environments and integrating it with electronic health records (EHR) and telemedicine platforms can improve reliability, adoption, and overall effectiveness.

In conclusion, these enhancements can transform the system into a more accurate, scalable, and user-centric digital healthcare solution, contributing significantly to improved dental diagnosis and preventive care.

REFERENCES

[1] L. Bussaban et al., "*Age Group Classification From Dental Panoramic Radiographs Using Deep Learning Techniques*," IEEE Access, vol. 12, pp. 139962–139973, 2024, doi: 10.1109/ACCESS.2024.3466953.

[2] S. B. Singh et al., "*Automatic Detection and Classification of Dental Anomalies and Tooth Types Using Transformer-Based YOLO With GA Optimization*," IEEE Access, vol. 13, pp. 1–15, 2025, doi: 10.1109/ACCESS.2025.3556523.

[3] M. Vijayalakshmi et al.,"*Advancing Oral Disease Diagnosis with Deep Learning and DenseNet Architecture*,"in Proc. 8th Int. Conf. on Electronics, Communication and Aerospace Technology (ICECA), 2024,doi: 10.1109/ICECA63461.2024.10801084.

[4] A. Sai Sachin and J. Karthikeyan, *"A Hybrid-Based Deep Learning Framework for Pill Detection, Multi-Attribute Classification, and Prediction With Metadata Retrieval,"*IEEE Access, vol. 14, pp. 1–18, 2026,doi:10.1109/ACCESS.2026.3654582.

[5] E. A. Olca*, "Professor X: Diagnosis and Treatment of Dermatological Diseases by Integration of Visual Diagnosis and Retrieval-Augmented Generation (RAG) Technologies,"*IEEEAccess,vol.13,pp.112,2025,doi:10.1109/ACCESS.2025.3636437.

[6] Agnes Paul Kattukaran, R.Chitra, Ashlin Santhosh et al., *"Dental Disease Detection Using Image Classification,"* Journal of Medical Imaging and Health Informatics, vol. 13, no. 2, pp. 450–458, 2023, doi: 10.1166/jmihi.2023.

[7] K. Moutselos, E. Berdouses, C. Oulis, *"Recognizing Occlusal Caries in Dental Intraoral Images Using Deep Learning,"* Computers in Biology and Medicine, vol. 152, pp. 106–115, 2023, doi: 10.1016/j.compbiomed.2023.106115.

[8] Athira Thulaseedharan ,et al.,*"Deep Learning-Based Detection of Oral Abnormalities in Clinical Images,"* Biomedical Signal Processing and Control, vol. 78, pp. 103–112, 2022, doi: 10.1016/j.bspc.2022.103112.

[9] Dr. M. Vijayalakshmi et al., *"CNN-Based Dental Disease Identification for Preventive Healthcare,"* Artificial Intelligence in Medicine, vol. 124, pp. 102–110, 2022, doi: 10.1016/j.artmed.2022.102110.

[10] Roshni Rathour et al., *"Dental Image Classification Using Deep Convolutional Neural Networks,"* IEEE Journal of Biomedical and Health Informatics, vol. 25, no. 6, pp. 2100–2108, 2021, doi: 10.1109/JBHI.2021.3052108.

[11] Radha Wasudeo Wande, Utkarsha Pacharaney et al*., "Dental Disease Detection Using EfficientNet,"*IEEE Access, vol. 11, pp. 125000–125010, 2023, doi: 10.1109/ACCESS.2023.

[12] L.R.Mehta, M. S. Borse et al., *"YOLOv7 for Dental Caries Detection: Challenges in Small Object Identification,"* Expert Systems with Applications, vol. 215, pp. 119–128, 2023, doi: 10.1016/j.eswa.2023.119128.

[13] Saiful Ulya, Muljono et al., *"A Comparative Study of YOLOv7 and YOLOv8 for Identification of Dental Caries on Intraoral RGB Images,"* Applied Sciences, vol. 14, no. 2, pp. 850–860, 2024, doi: 10.3390/app14020850.

[14] Duy-Linh Nguyen, Xuan-Thuy Vo, Adri Priadana et al., *"Improved YOLO-Based Model Using Attention and Extra Head Modules for Dental Disease Detection,"* IEEE Access, vol. 12, pp. 98000–98012, 2024, doi: 10.1109/ACCESS.2024.

[15] Çağatay Umut Öğdü, Kübra Arslanoğlu et al*,"Adaptive LLM-Based Clinical Decision Support System Integrating Biomedical RAG,"* IEEE Access, vol. 12, pp. 145000–145015, 2024, doi: 10.1109/ACCESS.2024.

[16] Mahizha, S. I., et al., *"Early Detection of Interproximal Caries using YOLOv8 Convolutional Neural Network on Bitewing Radiographs,"* Proc. ICDICI, 2024. doi: 10.1109/ICDICI62993.2024.10810960

[17] Nour, S. M., et al*., "Improving Automated Dental Diagnostics Utilizing YOLO9-Based Framework,"* Proc. ICCA, 2024,doi:10.1109/ICCA62237.2024.10927824

[18] Menon, G. A., et al., *"YOLO V5 Deep Learning Model for Dental Problem Detection,"* Proc. IC-RVITM, 2023,doi: 10.1109/IC-RVITM60032.2023.10434999

[19] Thulaseedharan, A., et al., "*Deep Learning Based Object Detection Algorithm for Detection of Dental Diseases and Differential Treatments,"* Proc. INDICON, 2022. doi: 10.1109/INDICON56171.2022.10040109

[20] Lal Priya P S, et al., "*Detection of Typical Pathologies and Differential Treatments in Dental Panoramic X-Rays based on Deep CNN,"* Proc. ICCC, 2023. DOI: 10.1109/ICCC57789.2023.10165614

APPENDIX

This appendix presents the implementation of the hybrid deep learning model that integrates YOLO for object detection and EfficientNet for classification. The system first detects regions of interest in dental X-ray images and then classifies the detected regions into specific dental conditions.

Stage 1: YOLO Model for Object Detection

In this stage, the YOLO model is loaded to detect regions such as cavities, implants, and impacted teeth from the input image.

```
from ultralytics import YOLO

# Load trained YOLO model
# YOLO trained weights
yolo_model = YOLO("best.pt")
```

Figure A.1 : YOLO Model

The YOLO model identifies bounding boxes around suspected dental anomalies. These detected regions are later used for further classification.

Stage 2: EfficientNet Model for Classification

This stage defines and loads the modified EfficientNet model used for classifying the detected regions.

```python
import torch
import torch.nn as nn
from torchvision import models

class ModifiedEfficientNet(nn.Module):
    def __init__(self, num_classes):
        super().__init__()
        self.backbone = models.efficientnet_b0(weights
            ="IMAGENET1K_V1")
        in_features = self.backbone.classifier[1].in_features
        self.backbone.classifier = nn.Identity()

        self.head = nn.Linear(in_features, num_classes)

    def forward(self, x):
        x = self.backbone(x)
        return self.head(x)

# Load trained EfficientNet weights
model = ModifiedEfficientNet(num_classes=4)
model.load_state_dict(torch.load("efficientnet.pth"))
model.eval()
```

Figure A.2 : EfficientNet Model

EfficientNet extracts deep features from cropped regions and classifies them into predefined dental categories such as cavity, filling, implant, or impacted tooth.

Stage 3: Image Preprocessing

Before classification, detected regions must be resized and converted into tensors compatible with the EfficientNet model.

```python
from torchvision import transforms
from PIL import Image
transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor()
])
```

Figure A.3 : Preprocessing

This preprocessing step ensures uniform input size and normalization, which is required for consistent model performance.

Stage 4: Integrated Detection and Classification Pipeline

This stage combines YOLO detection and EfficientNet classification into a single pipeline.

```
def detect_and_classify(image_path):
    img = Image.open(image_path).convert("RGB")
    # YOLO detection
    results = yolo_model(image_path)
    final_results = []
    for box in results[0].boxes:
        x1, y1, x2, y2 = map(int, box.xyxy[0])
        # Crop detected region
        cropped = img.crop((x1, y1, x2, y2))
        # Preprocess
        input_tensor = transform(cropped).unsqueeze(0)
        # Classification
        with torch.no_grad():
            output = model(input_tensor)
            pred = torch.argmax(output, dim=1).item()
        final_results.append({
            "Bounding Box": [x1, y1, x2, y2],
            "Predicted Class": pred
        })

    return final_results
```

Figure A.4 : Integrated Pipeline

The YOLO model first detects regions of interest. Each region is cropped and passed to EfficientNet, which assigns a class label. This combined approach improves both localization and classification accuracy.

Stage 5: RAG-Based Recommendation System

This stage generates personalized dental recommendations based on the detected conditions from the YOLO + EfficientNet model and user lifestyle inputs. The system uses a Retrieval-Augmented Generation (RAG) approach, where relevant knowledge is retrieved from a structured database and processed through rule-based logic.

```
from rag_engine.retriever import retrieve_conditions
from rag_engine.engine import apply_rules
def generate_ai_output(detections, lifestyle):
    # Retrieve knowledge base data based on detected conditions
    kb_data = retrieve_conditions(detections)
    # Apply rule-based logic to generate recommendations
    result = apply_rules(detections, lifestyle, kb_data)
    return result
```

Figure A.5 : RAG System

The RAG-based module enhances the system by transforming detection results into actionable, user-specific insights, making the overall solution more practical and patient-centric.

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The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.